

Unit 10. Autoregression and Forecasting

1 Autoregressive Models

FREES §8.2, §8.5.1.

Unit 9 left the dependence between y_t and its past as a diagnostic finding: the correlogram of a regression's residuals was read for departures from white noise, and the remedy offered was differencing. Modelling that dependence directly means putting the past of the series on the right of the equation.

1.1 The first-order process

Definition 10.1 (White noise and the innovations). Throughout, $\{\varepsilon_t\}_{t \in \mathbb{Z}}$ is independent and identically distributed with mean 0 and variance $\sigma^2 \in (0, \infty)$, and $\mathcal{L}_t$ denotes the past space of the prerequisites: the closure in L^2 of the square-integrable functions of finitely many ε_u with $u \leq t$.

Definition 10.2 (Autoregressive model of order one). The $AR(1)$ model is

$$y_t = \beta_0 + \beta_1 y_{t-1} + \varepsilon_t, \tag{1}$$

with $|\beta_1| < 1$ and ε_t independent of $\mathcal{L}_{t-1}$.

Theorem 10.3 (Existence, uniqueness, moments). Let $|\beta_1| < 1$ and $\mu = \beta_0/(1 - \beta_1)$. Then

$$y_t = \mu + \sum_{k=0}^{\infty} \beta_1^k \varepsilon_{t-k} \quad (2)$$

converges in L^2 , satisfies (1), and is the only weakly stationary solution of (1). It is weakly stationary with

$$\mathbb{E}[y_t] = \mu, \quad \gamma(0) = \text{var}(y_t) = \frac{\sigma^2}{1 - \beta_1^2}, \quad \gamma(k) = \beta_1^k \gamma(0), \quad \rho_k = \beta_1^k. \quad (3)$$

Proof. The coefficients β_1^k are square summable, so (2) converges in L^2 and lies in $\mathcal{L}_t$ by the prerequisites' construction of the past space. Splitting off the first term,

$$y_t = \mu + \varepsilon_t + \beta_1 \sum_{k=0}^{\infty} \beta_1^k \varepsilon_{t-1-k} = \mu + \varepsilon_t + \beta_1(y_{t-1} - \mu),$$

and $\mu - \beta_1\mu = \beta_0$, so (1) holds.

For the moments, $\mathbf{E}[y_t] = \mu$ because each ε has mean zero and L^2 convergence preserves the mean. The innovations are uncorrelated with common variance σ^2 , so for $k \geq 0$

$$\gamma(k) = \text{cov}\left(\sum_{j \geq 0} \beta_1^j \varepsilon_{t-j}, \sum_{l \geq 0} \beta_1^l \varepsilon_{t+k-l}\right) = \sigma^2 \sum_{j \geq 0} \beta_1^j \beta_1^{j+k} = \sigma^2 \beta_1^k \sum_{j \geq 0} \beta_1^{2j} = \frac{\sigma^2 \beta_1^k}{1 - \beta_1^2},$$

the pairing $j = l - k$ being the only one that survives. Both moments are free of t .

For uniqueness, let $\{\tilde{y}_t\}$ be another weakly stationary solution and put $d_t = y_t - \tilde{y}_t$. Subtracting the two copies of (1) gives $d_t = \beta_1 d_{t-1} = \beta_1^m d_{t-m}$ for every m . Weak stationarity bounds $\mathbf{E}[d_{t-m}^2]$ by a constant C free of m , so $\mathbf{E}[d_t^2] \leq \beta_1^{2m} C \rightarrow 0$, forcing $d_t = 0$ almost surely. $\square$

Remark. The autocorrelation function decays geometrically and never reaches zero. That is the identifying signature of an $AR(1)$: a correlogram falling off at a constant ratio, against the near-constant correlogram of a random walk (Unit 9) and the immediate collapse to zero of white noise.

Proposition 10.4 (The two boundary cases). Setting $\beta_1 = 0$ in (1) gives white noise about β_0 . Setting $\beta_1 = 1$ gives $\Delta y_t = \beta_0 + \varepsilon_t$, a random walk with drift β_0 , for which no weakly stationary solution exists.

Proof. The first is immediate. For the second, $\beta_1 = 1$ turns (1) into $y_t - y_{t-1} = \beta_0 + \varepsilon_t$, which is Unit 9's random walk. If a weakly stationary solution existed, taking variances in $y_t = y_{t-1} + \beta_0 + \varepsilon_t$ would give $\gamma(0) = \gamma(0) + \sigma^2$, since ε_t is uncorrelated with y_{t-1} , and $\sigma^2 > 0$. $\square$

Remark. The variance identity $\gamma(0)(1 - \beta_1^2) = \sigma^2$ of (3) says the series is more variable than its innovations, and increasingly so as β_1 approaches 1. In cross-sectional regression the regressors are fixed and the response inherits the error variance exactly; here the regressor is the response one step earlier and the variance accumulates.

Proposition 10.5 (Fourth moment). If in addition $\mu_4 = \mathbf{E}[\varepsilon_t^4] < \infty$, then $x_t = y_t - \mu$ satisfies

$$\mathbf{E}[x_t^4] = \frac{\mu_4 + 6\beta_1^2\sigma^2\gamma(0)}{1 - \beta_1^4} < \infty, \quad (4)$$

and $\{x_t^2\}$ is weakly stationary with autocovariance $\text{cov}(x_t^2, x_{t+k}^2) = \beta_1^{2k} \text{var}(x_t^2) \rightarrow 0$. Consequently $n^{-1} \sum_{t \leq n} x_t^2 \xrightarrow{p} \gamma(0)$.

Proof. Finiteness. For the partial sum $S_m = \sum_{j < m} \beta_1^j \varepsilon_{t-j}$, expanding the fourth power and using independence with $\mathbf{E}[\varepsilon] = 0$ leaves only the terms in which every index occurs at least twice: the diagonal, and the three ways of splitting four factors into two distinct pairs. Hence

$$\mathbf{E}[S_m^4] = \mu_4 \sum_{j < m} \beta_1^{4j} + 3\sigma^4 \left\{ \left(\sum_{j < m} \beta_1^{2j} \right)^2 - \sum_{j < m} \beta_1^{4j} \right\} \leq \frac{\mu_4}{1 - \beta_1^4} + \frac{3\sigma^4}{(1 - \beta_1^2)^2},$$

a bound free of m . Now $S_m \rightarrow x_t$ in L^2 , hence in probability and so almost surely along a subsequence, and Fatou's lemma gives $\mathbf{E}[x_t^4] < \infty$. The law of x_t is free of t , being the weak limit of the laws of the S_m , which are.

Value. Write $m_4 = \mathbf{E}[x_t^4]$ and use $x_t = \beta_1 x_{t-1} + \varepsilon_t$ with ε_t independent of x_{t-1} . All four cross terms in the expansion carry a factor $\mathbf{E}[\varepsilon_t] = 0$ or $\mathbf{E}[\varepsilon_t^3] \mathbf{E}[x_{t-1}] = 0$, so $m_4 = \beta_1^4 m_4 + 6\beta_1^2 \sigma^2 \mathbf{E}[x_{t-1}^2] + \mu_4$, which is (4).

Autocovariance. Iterate (1) k times to get $x_{t+k} = \beta_1^k x_t + v$ with $v = \sum_{j < k} \beta_1^j \varepsilon_{t+k-j}$ independent of x_t and of mean zero. Then $x_{t+k}^2 = \beta_1^{2k} x_t^2 + 2\beta_1^k x_t v + v^2$, and $\text{cov}(x_t^2, x_t v) = \mathbf{E}[x_t^3] \mathbf{E}[v] - \mathbf{E}[x_t^2] \mathbf{E}[x_t] \mathbf{E}[v] = 0$ while $\text{cov}(x_t^2, v^2) = 0$, both by independence. So $\text{cov}(x_t^2, x_{t+k}^2) = \beta_1^{2k} \text{var}(x_t^2)$, which tends to 0, and the stationary law of large numbers of the prerequisites applies. $\square$

1.2 Higher orders and moving averages

Definition 10.6 (Backshift). $\mathbf{B}y_t = y_{t-1}$ and $\mathbf{B}^k y_t = y_{t-k}$. For a polynomial $\Phi(x) = 1 - \beta_1 x - \cdots - \beta_p x^p$, $\Phi(\mathbf{B})y_t = y_t - \beta_1 y_{t-1} - \cdots - \beta_p y_{t-p}$.

Definition 10.7 ($AR(p)$, $MA(q)$, $ARMA$, $ARIMA$). With $\Phi(x) = 1 - \beta_1 x - \cdots - \beta_p x^p$ and $\Theta(x) = 1 - \theta_1 x - \cdots - \theta_q x^q$,

$$AR(p) : \quad \Phi(\mathbf{B})y_t = \beta_0 + \varepsilon_t, \quad (5)$$

$$MA(q) : \quad y_t = \beta_0 + \Theta(\mathbf{B})\varepsilon_t, \quad (6)$$

$$ARMA(p, q) : \quad \Phi(\mathbf{B})y_t = \beta_0 + \Theta(\mathbf{B})\varepsilon_t, \quad (7)$$

and $ARIMA(p, d, q)$ is $ARMA(p, q)$ applied to $w_t = (1 - \mathbf{B})^d y_t$.

Proposition 10.8 ($MA(q)$ is always stationary and its correlogram cuts off). The $MA(q)$ process (6) is weakly stationary for every $\theta_1, \dots, \theta_q$, with $\theta_0 = -1$ and

$$\gamma(k) = \sigma^2 \sum_{j=0}^{q-k} \theta_j \theta_{j+k} \quad (0 \leq k \leq q), \quad \gamma(k) = 0 \quad (k > q). \quad (8)$$

Proof. $y_t - \beta_0$ is a finite linear combination of $\varepsilon_t, \dots, \varepsilon_{t-q}$ with coefficients $-\theta_j$ and $\theta_0 = -1$, so the mean is β_0 and the covariance is $\sigma^2 \sum_j \theta_j \theta_{j+k}$ over those j with both indices in $\{0, \dots, q\}$, all free of t . For $k > q$ the two windows $\{t - q, \dots, t\}$ and $\{t + k - q, \dots, t + k\}$ are disjoint and the innovations independent. $\square$

Theorem 10.9 (Stationarity of $AR(p)$). Suppose Φ has p distinct roots $g_1, \dots, g_p$, all with $|g_i| > 1$. Then $1/\Phi(x) = \sum_{k \geq 0} \psi_k x^k$ for $|x| < \min_i |g_i|$, with

$$\psi_k = \sum_{i=1}^p c_i g_i^{-k}, \quad c_i = \prod_{l \neq i} (1 - g_i/g_l)^{-1}, \quad (9)$$

and $|\psi_k| \leq (\sum_i |c_i|) \varrho^k$ with $\varrho = \max_i |g_i|^{-1} < 1$. Consequently $y_t = \mu + \sum_{k \geq 0} \psi_k \varepsilon_{t-k}$, with $\mu = \beta_0 / \Phi(1)$, converges in L^2 and is the unique weakly stationary solution of (5).

Proof. Since the roots are distinct, $\Phi(x) = \prod_i (1 - x/g_i)$ and the partial fraction expansion $1/\Phi(x) = \sum_i c_i / (1 - x/g_i)$ holds with c_i obtained by multiplying through by $(1 - x/g_i)$ and setting $x = g_i$, which gives $c_i = \prod_{l \neq i} (1 - g_i/g_l)^{-1}$. Expanding each factor as a geometric series, valid for $|x| < |g_i|$, and collecting powers gives (9) and the bound.

Absolute summability of $\{\psi_k\}$ makes the series square summable, so it converges in L^2 . Applying $\Phi(B)$ term by term, the coefficient of ε_{t-k} in $\Phi(B) \sum_j \psi_j \varepsilon_{t-j}$ is the k th coefficient of $\Phi(x) \sum_j \psi_j x^j = 1$, hence 1 for $k = 0$ and 0 otherwise; the interchange is legitimate because $\Phi(B)$ is a finite linear combination and L^2 limits are preserved by such. So $\Phi(B)(y_t - \mu) = \varepsilon_t$, and $\Phi(B)\mu = \mu\Phi(1) = \beta_0$.

For uniqueness, let d_t be the difference of two weakly stationary solutions, so $\Phi(B)d_t = 0$ and $E[d_t^2] \leq C$ free of t . Multiplying the truncated series $\sum_{k \leq K} \psi_k x^k$ by $\Phi(x)$ leaves 1 plus terms in $x^{K+1}, \dots, x^{K+p}$ whose coefficients are sums of at most p products $\psi_k \beta_j$ with $k > K$, hence

bounded by $p \max_j |\beta_j| \sum_i |c_i| \varrho^K$. Applying the same identity with B in place of x to d_t ,

$$0 = \sum_{k \leq K} \psi_k \Phi(B) d_{t-k} = d_t + \sum_{j=K+1}^{K+p} e_{K,j} d_{t-j}, \quad |e_{K,j}| \leq C' \varrho^K,$$

so $\mathbf{E}[d_t^2]^{1/2} \leq pC' \varrho^K C^{1/2} \rightarrow 0$ as $K \rightarrow \infty$ by the triangle inequality in L^2 , forcing $d_t = 0$ almost surely. $\square$

Remark. For $p = 1$ the condition is $|g_1| = |\beta_1|^{-1} > 1$, that is $|\beta_1| < 1$. The requirement is on the roots, not on the coefficients: an $AR(2)$ with $\beta_1 = 1.02$ and $\beta_2 = -0.24$ is stationary, because $\Phi(x) = 1 - 1.02x + 0.24x^2$ has both roots outside the unit circle, while its first coefficient exceeds one.

Definition 10.10 (ψ -coefficients). The representation $y_t = \mu + \sum_{k \geq 0} \psi_k \varepsilon_{t-k}$ with $\psi_0 = 1$ is the **ψ -coefficient representation**. For $AR(1)$, $\psi_k = \beta_1^k$; for $MA(q)$, $\psi_k = -\theta_k$ for $1 \leq k \leq q$ and 0 beyond.

1.3 Partial autocorrelation

Definition 10.11 (Partial autocorrelation). For a weakly stationary series, let $(\alpha_{1,k}, \dots, \alpha_{k,k})$ be the coefficients of the projection of $y_t - \mu$ on $\{y_{t-1} - \mu, \dots, y_{t-k} - \mu\}$, that is the minimiser of $\mathbf{E}[\{y_t - \mu - \sum_{j \leq k} a_j (y_{t-j} - \mu)\}^2]$. The **partial autocorrelation** at lag k is $\alpha_{k,k}$.

Theorem 10.12 (The partial correlogram of an $AR(p)$ cuts off). Let $\{y_t\}$ be the stationary $AR(p)$ of Theorem 10.9. Then $(\alpha_{1,k}, \dots, \alpha_{k,k}) = (\beta_1, \dots, \beta_p, 0, \dots, 0)$ for every $k \geq p$. In particular $\alpha_{p,p} = \beta_p$ and $\alpha_{k,k} = 0$ for $k > p$.

Proof. Write $x_t = y_t - \mu$, so $x_t = \sum_{j \leq p} \beta_j x_{t-j} + \varepsilon_t$. Fix $k \geq p$ and let $a = (\beta_1, \dots, \beta_p, 0, \dots, 0)$. The residual of that candidate is exactly ε_t , and $x_{t-j} \in \mathcal{L}_{t-1}$ for $j \geq 1$ while ε_t is independent of $\mathcal{L}_{t-1}$ with mean zero, so $\mathbf{E}[\varepsilon_t x_{t-j}] = 0$ for $j = 1, \dots, k$. Those are the normal equations of the projection (the projection theorem of the prerequisites), whose solution is unique when the covariance matrix of $(x_{t-1}, \dots, x_{t-k})$ is nonsingular. It is: a singular covariance matrix would make some combination $\sum_{j \leq k} b_j x_{t-j}$ with $b \neq \mathbf{0}$ vanish almost surely. Since $\text{cov}(x_{t-j}, \varepsilon_{t-l}) = \sigma^2 \psi_{l-j}$, which is 0 for $l < j$ and σ^2 for $l = j$, taking the covariance of that combination with ε_{t-1} gives

$b_1\sigma^2 = 0$, then with ε_{t-2} gives $b_2\sigma^2 = 0$, and so on: the system is triangular and forces every $b_j = 0$. $\square$

Remark. The correlogram and the partial correlogram are complementary identifiers: an $MA(q)$ has $\rho_k = 0$ beyond lag q and a partial correlogram that decays without cutting off, an $AR(p)$ has $\alpha_{k,k} = 0$ beyond lag p and a correlogram that decays without cutting off. Neither cuts off for an $ARMA$ with both orders positive.

Proposition 10.13 (Yule–Walker equations). The coefficients of Definition 10.11 solve the linear system

$$\rho_l = \sum_{j=1}^k \alpha_{j,k} \rho_{|l-j|}, \quad l = 1, \dots, k. \quad (10)$$

Proof. The normal equations of the projection in Definition 10.11 are $\mathbf{E}[\{x_t - \sum_j a_j x_{t-j}\} x_{t-l}] = 0$ for $l = 1, \dots, k$, where $x_t = y_t - \mu$; that is $\gamma(l) = \sum_j a_j \gamma(l-j)$, and dividing by $\gamma(0)$ gives (10). $\square$

Remark. Two sample versions of $\hat{\alpha}_{k,k}$ are in use and they are not the same number. Substituting the sample autocorrelations r_k into (10) and solving, which is what `pacf` does, forms every inner product from all T observations; regressing y_t on an intercept and $y_{t-1}, \dots, y_{t-k}$ forms them from the $T - k$ dates at which every regressor exists. The difference is an end effect of order T^{-1} , and §5.1 both reproduces the package exactly from (10) and measures the gap between the two.

2 Estimation, Diagnostics, Forecasting

FREES §8.3, §8.4, §8.6.

2.1 Conditional least squares

Definition 10.14 (Conditional least squares). Given $y_1, \dots, y_T$, the **conditional least squares** estimates of β_0, β_1 in (1) minimise $\sum_{t=2}^T (y_t - b_0 - b_1 y_{t-1})^2$: ordinary least squares of y_t on y_{t-1} over the $T - 1$ usable dates. The residuals are $e_t = y_t - \hat{\beta}_0 - \hat{\beta}_1 y_{t-1}$ and

$$s^2 = \frac{1}{T - 3} \sum_{t=2}^T (e_t - \bar{e})^2. \quad (11)$$

Remark. The word **conditional** records that the first observation contributes no term: the criterion is the sum of squared one-step errors given y_1 , not the likelihood of the whole series. The divisor in (11) is the number of usable dates $T - 1$ minus the two fitted parameters, and $\bar{e}$ appears because the residuals of a regression that omits one observation need not sum to zero.

Proposition 10.15 (Relation to the sample autocorrelation). Write $\bar{y}_- = (T - 1)^{-1} \sum_{t=1}^{T-1} y_t$

and $\bar{y}_+ = (T - 1)^{-1} \sum_{t=2}^T y_t$. Then

$$\hat{\beta}_1 = \frac{\sum_{t=2}^T (y_{t-1} - \bar{y}_-)(y_t - \bar{y}_+)}{\sum_{t=2}^T (y_{t-1} - \bar{y}_-)^2}, \quad \hat{\beta}_0 = \bar{y}_+ - \hat{\beta}_1 \bar{y}_-, \quad (12)$$

and $\hat{\beta}_1 - r_1 = O(1/T)$ when no observation dominates, r_1 being the lag one sample autocorrelation of Unit 9.

Proof. The displayed formulas are the $k = 1$ least squares expressions of Unit 2 applied to the $T - 1$ pairs (y_{t-1}, y_t) . Comparing with r_1 , whose numerator centres both factors at the overall $\bar{y}$ and whose denominator runs over all T observations, the numerators differ by terms involving $\bar{y}_\pm - \bar{y} = O(1/T)$ and the denominators by the single term $(y_T - \bar{y})^2$. $\square$

2.2 Large-sample theory

The summands of the numerator in (12) are $y_{t-1}\varepsilon_t$, which are uncorrelated but not independent, so the central limit theorems of Units 7 and 8 do not apply to them. The prerequisites' central limit theorem for autoregressive scores does.

Theorem 10.16 (Asymptotics of conditional least squares). Let $\{y_t\}$ be the stationary $AR(1)$ of Theorem 10.3 with $\mathbf{E}[\varepsilon_t^4] < \infty$. Then, as $T \rightarrow \infty$,

$$\hat{\beta}_1 \xrightarrow{p} \beta_1, \quad \sqrt{T}(\hat{\beta}_1 - \beta_1) \xrightarrow{d} N(0, 1 - \beta_1^2). \quad (13)$$

Proof. Write $x_t = y_t - \mu$, so that (12) becomes, after subtracting μ from every observation,

$$\hat{\beta}_1 - \beta_1 = \frac{\sum_{t=2}^T (x_{t-1} - \bar{x}_-) \varepsilon_t}{\sum_{t=2}^T (x_{t-1} - \bar{x}_-)^2},$$

because $x_t = \beta_1 x_{t-1} + \varepsilon_t$ makes the numerator of (12) equal to $\beta_1 \sum (x_{t-1} - \bar{x}_-)^2 + \sum (x_{t-1} - \bar{x}_-) \varepsilon_t$, the centring of the second factor contributing nothing since $\sum_t (x_{t-1} - \bar{x}_-) = 0$.

Denominator. $\{x_t\}$ is weakly stationary with $\gamma(k) = \beta_1^k \gamma(0) \rightarrow 0$, so $\bar{x}_- \xrightarrow{p} 0$ by the stationary law of large numbers of the prerequisites, and $\{x_t^2\}$ is weakly stationary with vanishing

autocovariance by Proposition 10.5, so $T^{-1} \sum_t x_{t-1}^2 \xrightarrow{p} \gamma(0)$. Hence

$$\frac{1}{T} \sum_{t=2}^T (x_{t-1} - \bar{x}_-)^2 = \frac{1}{T} \sum_{t=2}^T x_{t-1}^2 - \frac{T-1}{T} \bar{x}_-^2 \xrightarrow{p} \gamma(0) > 0.$$

Numerator. With $a_j = \beta_1^j$ the sequence $x_{t-1} = \sum_{j \geq 0} a_j \varepsilon_{t-1-j}$ is exactly the linear combination of the prerequisites' central limit theorem for autoregressive scores, whose limit variance is $\sigma^2 \cdot \sigma^2 \sum_j \beta_1^{2j} = \sigma^2 \gamma(0)$. So $T^{-1/2} \sum_t x_{t-1} \varepsilon_t \xrightarrow{d} N(0, \sigma^2 \gamma(0))$. The centring costs nothing:

$$\frac{1}{\sqrt{T}} \sum_{t=2}^T (x_{t-1} - \bar{x}_-) \varepsilon_t = \frac{1}{\sqrt{T}} \sum_{t=2}^T x_{t-1} \varepsilon_t - \bar{x}_- \cdot \frac{1}{\sqrt{T}} \sum_{t=2}^T \varepsilon_t,$$

and the second term is a product of $\bar{x}_- \xrightarrow{p} 0$ with a quantity converging in distribution to $N(0, \sigma^2)$ by the prerequisites' central limit theorem for independent summands, hence tends to 0 in probability by Slutsky's theorem.

Ratio. Slutsky's theorem again gives $\sqrt{T}(\hat{\beta}_1 - \beta_1) \xrightarrow{d} N(0, \sigma^2 \gamma(0) / \gamma(0)^2)$, and $\sigma^2 / \gamma(0) = 1 - \beta_1^2$ by (3). Consistency follows from the same two limits with the $\sqrt{T}$ removed. $\square$

Corollary 10.17 (Standard error). An approximate standard error for $\hat{\beta}_1$ is $\sqrt{(1 - \hat{\beta}_1^2)/T}$, and it agrees to order T^{-1} with the ordinary least squares standard error $s/\{\sum_t (y_{t-1} - \bar{y}_-)^2\}^{1/2}$ reported by a regression routine.

Proof. The first expression is (13) with β_1 estimated. For the second, the regression standard error squared is $s^2/\sum_t (y_{t-1} - \bar{y}_-)^2$, whose numerator converges in probability to σ^2 and whose denominator is T times a quantity converging to $\gamma(0)$; the ratio times T therefore tends to $\sigma^2/\gamma(0) = 1 - \beta_1^2$. $\square$

Remark. Two facts are worth separating. The reported standard error is asymptotically correct, so the usual t ratio may be used: the dependence in the design does not invalidate the regression output the way Unit 9's random walk did. But it is correct only asymptotically, and $\hat{\beta}_1$ is biased downwards in finite samples, the more so as β_1 approaches 1; §5.2 measures both.

Remark. The precision of $\hat{\beta}_1$ improves as $|\beta_1|$ grows: the asymptotic variance $(1 - \beta_1^2)/T$ vanishes at the boundary. That is the first sign that the boundary is a different regime, and §3.4 shows the

whole distribution changes there.

2.3 Residual diagnostics

Unit 9 measured the coverage of the $\pm 2/\sqrt{T}$ correlogram band by simulation, because the summands $\varepsilon_t \varepsilon_{t-k}$ of the numerator of r_k are dependent and no central limit theorem then available reached them. Writing them as $u_{t-1} \varepsilon_t$ with $u_{t-1} = \varepsilon_{t-k}$ puts them in the form the prerequisites cover.

Theorem 10.18 (Joint limit of the sample autocorrelations). Let $y_1, \dots, y_T$ be white noise with mean μ and variance σ^2 , and fix $K \geq 1$. Then

$$\sqrt{T} (r_1, \dots, r_K)' \xrightarrow{d} N(\mathbf{0}, \mathbf{I}_K). \quad (14)$$

Proof. Put $\xi_t = y_t - \mu$ and $\mathbf{a}_j = \mathbf{e}_{j+1} \in \mathbb{R}^K$ for $0 \leq j \leq K-1$, $\mathbf{a}_j = \mathbf{0}$ beyond, so that $\mathbf{u}_{t-1} = \sum_j \mathbf{a}_j \xi_{t-1-j} = (\xi_{t-1}, \dots, \xi_{t-K})'$ and $\mathbf{\Gamma} = \mathbf{E}[\mathbf{u}\mathbf{u}'] = \sigma^2 \mathbf{I}_K$. The prerequisites' theorem gives

$$\frac{1}{\sqrt{T}} \sum_{t=K+1}^T \mathbf{u}_{t-1} \xi_t \xrightarrow{d} N(\mathbf{0}, \sigma^4 \mathbf{I}_K),$$

the theorem's sum running from $t = 1$ and this one from $t = K + 1$, a difference of K terms each bounded in probability and multiplied by $T^{-1/2}$.

The k th coordinate of the sum is $\sum_t \xi_t \xi_{t-k}$, the numerator of r_k with the mean known. Writing $\bar{\xi} = \bar{y} - \mu$, the numerator actually used differs from it by

$$-\bar{\xi} \sum_t \xi_{t-k} - \bar{\xi} \sum_t \xi_t + (T - k)\bar{\xi}^2,$$

and $\bar{\xi} = O_p(T^{-1/2})$ while each inner sum is $O_p(T^{1/2})$, so all three terms are $O_p(1)$ and vanish after division by $\sqrt{T}$. The denominator satisfies $T^{-1} \sum_t (y_t - \bar{y})^2 \xrightarrow{p} \sigma^2$ by the weak law. Dividing and applying Slutsky's theorem gives (14). $\square$

Corollary 10.19 (The correlogram band). Under white noise, $\mathbf{P}\{|r_k| > 2/\sqrt{T}\} \rightarrow \mathbf{P}\{|Z| > 2\} = 0.0455$ for each fixed k , Z being standard normal.

Proof. $\sqrt{T} r_k \xrightarrow{d} N(0, 1)$ by Theorem 10.18, and ± 2 are continuity points of the normal distribution function. $\square$

Remark. Unit 9 could prove only the Chebyshev bound $\frac{1}{4}$ and measured 95.6% and 95.8% coverage by simulation. Those two numbers are now explained: the band is a 95.45% band in the limit.

Definition 10.20 (Portmanteau statistics). For residual autocorrelations $r_k(e)$ and a user-chosen K ,

$$Q_{BP} = T \sum_{k=1}^K r_k(e)^2, \quad Q_{LB} = T(T+2) \sum_{k=1}^K \frac{r_k(e)^2}{T-k}. \quad (15)$$

Theorem 10.21 (Limit for an unfitted series). If $y_1, \dots, y_T$ is white noise and the r_k are computed from the series itself, then $Q_{BP} \xrightarrow{d} \chi_K^2$ and $Q_{LB} \xrightarrow{d} \chi_K^2$.

Proof. $Q_{BP} = |\sqrt{T} \mathbf{r}|^2$ with $\mathbf{r} = (r_1, \dots, r_K)'$, and Theorem 10.18 with the continuous mapping

theorem gives $|\sqrt{T}\mathbf{r}|^2 \xrightarrow{d} |Z|^2$ for $Z \sim N(\mathbf{0}, \mathbf{I}_K)$, which is χ_K^2 by the prerequisites. For Q_{LB} , write

$$Q_{LB} = \sum_{k=1}^K \frac{T(T+2)}{T-k} r_k^2 = \sum_{k=1}^K \left(1 + \frac{k+2}{T-k}\right) T r_k^2,$$

so $Q_{LB} - Q_{BP} = \sum_k \{(k+2)/(T-k)\} T r_k^2 \xrightarrow{p} 0$, each $T r_k^2$ being bounded in probability and each factor $O(1/T)$; Slutsky's theorem finishes. $\square$

Remark. When the r_k are computed from the residuals of a fitted $ARMA(p, q)$, the limit is χ_{K-p-q}^2 rather than χ_K^2 : fitting removes exactly $p + q$ directions from the residual correlogram. That statement is **not** proved here. It needs the joint limit of the residual autocorrelations, which involves the estimation error and is beyond the prerequisites as they stand. §5.3 measures it instead, and finds the mean of Q_{LB} at $K - 1$ and the size at the nominal 5% only when the degrees of freedom are reduced.

Remark. Q_{LB} inflates each term by $(T + 2)/(T - k)$, a correction that matters only for k comparable with T . Its usual justification is that $\text{var}(r_k)$ is closer to $(T - k)/\{T(T + 2)\}$ than to $1/T$ in small samples; the limit theorem above cannot distinguish the two.

2.4 Forecasting

Definition 10.22 (Chain rule). The l -step forecast from an $AR(1)$ fit is defined recursively by $\hat{y}_{T+l} = \hat{\beta}_0 + \hat{\beta}_1 \hat{y}_{T+l-1}$, starting from $\hat{y}_T = y_T$.

Theorem 10.23 (Optimal forecast and its error). For the stationary $AR(1)$ with known parameters, the minimum mean squared error predictor of y_{T+l} among all square-integrable functions of the past is

$$\mathbb{E}[y_{T+l} \mid \mathcal{L}_T] = \mu + \beta_1^l (y_T - \mu), \quad (16)$$

and its error variance is

$$\text{var}(y_{T+l} - \hat{y}_{T+l}) = \sigma^2 \sum_{k=0}^{l-1} \beta_1^{2k} = \sigma^2 \frac{1 - \beta_1^{2l}}{1 - \beta_1^2}. \quad (17)$$

The point forecast tends to μ and the error variance to $\gamma(0)$ as $l \rightarrow \infty$.

Proof. Iterating (1) forward l steps,

$$y_{T+l} - \mu = \beta_1^l (y_T - \mu) + \sum_{k=0}^{l-1} \beta_1^k \varepsilon_{T+l-k}. \quad (18)$$

Each innovation in the sum has index exceeding T , hence is independent of $\mathcal{L}_T$ and has mean zero, while $y_T - \mu \in \mathcal{L}_T$; taking conditional expectations gives (16), which minimises mean squared error by the projection property of conditional expectation in L^2 (prerequisites). The forecast error is the sum in (18), whose terms are uncorrelated with variances $\sigma^2 \beta_1^{2k}$, giving (17). The limits follow from $|\beta_1| < 1$ and (3). $\square$

Corollary 10.24 (Forecast interval). An approximate $100(1 - \alpha)\%$ interval for y_{T+l} is

$$\hat{y}_{T+l} \pm t_{T-3, 1-\alpha/2} s \left(\sum_{k=0}^{l-1} \hat{\beta}_1^{2k} \right)^{1/2}, \quad (19)$$

which for $l = 1$ is $\hat{y}_{T+1} \pm t s$ and for $l = 2$ is $\hat{y}_{T+2} \pm t s(1 + \hat{\beta}_1^2)^{1/2}$.

Proof. Substitute the estimates into (17) and treat the estimation error in $\hat{\beta}_0, \hat{\beta}_1$ as negligible, which Theorem 10.16 justifies to order $T^{-1/2}$ against the $O(1)$ term σ . $\square$

Theorem 10.25 (General ψ -weight forecast). For any process with the representation $y_t = \mu + \sum_{k \geq 0} \psi_k \varepsilon_{t-k}$ and $\sum_k |\psi_k| < \infty$,

$$\mathbb{E}[y_{T+l} \mid \mathcal{L}_T] = \mu + \sum_{k \geq l} \psi_k \varepsilon_{T+l-k}, \quad \text{var}(y_{T+l} - \mathbb{E}[y_{T+l} \mid \mathcal{L}_T]) = \sigma^2 \sum_{k=0}^{l-1} \psi_k^2. \quad (20)$$

Proof. Split the sum at $k = l$. The terms with $k < l$ carry innovations dated after T , which are independent of $\mathcal{L}_T$ with mean zero; the terms with $k \geq l$ carry innovations dated at or before T , which lie in $\mathcal{L}_T$. The forecast error is therefore $\sum_{k < l} \psi_k \varepsilon_{T+l-k}$, a sum of uncorrelated terms. $\square$

Remark. Three forecast-interval shapes have now been derived, and they differ in what the width does with the horizon. A linear trend with white noise errors (Unit 9) gives almost constant width; a random walk gives width growing like $\sqrt{t}$; a stationary $AR(1)$ gives width increasing to the finite limit $\sqrt{\gamma(0)}$, which is the standard deviation of the series itself. The last is the statement that beyond a few multiples of the correlation time nothing is known about the future except the marginal distribution.

Definition 10.26 (Smoothed series). The fitted values $\hat{y}_t = \hat{\beta}_0 + \hat{\beta}_1 y_{t-1}$, $t = 2, \dots, T$, are the **smoothed series**: the one-step forecast made at each date from the estimated model.

3 Smoothing, Seasonality, Unit Roots, Volatility

FREES §9.1–9.5.

3.1 Moving averages

Definition 10.27 (Running average). The running average of length k is $\hat{s}_t = k^{-1} \sum_{j=0}^{k-1} y_{t-j}$, and the forecast it supplies is $\hat{y}_{T+l} = \hat{s}_T$ for every $l \geq 1$.

Proposition 10.28 (Update and weighted least squares). (i) $\hat{s}_t = \hat{s}_{t-1} + (y_t - y_{t-k})/k$.
(ii) $\hat{s}_T$ is the weighted least squares estimate of β_0 in $y_t = \beta_0 + \varepsilon_t$ with weights $w_t = \mathbf{1}\{t > T - k\}$.
(iii) If $\{\varepsilon_t\}$ is white noise then $\text{var}(\hat{s}_t) = \sigma^2/k$.

Proof. (i) $k\hat{s}_t - k\hat{s}_{t-1} = y_t - y_{t-k}$, the two windows differing in one term at each end. (ii) The weighted criterion is $\sum_{t>T-k} (y_t - b_0)^2$, minimised at the mean of the last k observations (Unit 4). (iii) The average of k uncorrelated terms. $\square$

Proposition 10.29 (A running average lags a linear trend). If $y_t = \beta_0 + \beta_1 t + \varepsilon_t$ then

$$\mathbb{E}[\hat{s}_t] = \beta_0 + \beta_1 \left(t - \frac{k-1}{2}\right), \quad \mathbb{E}[\hat{s}_t^{(2)}] = \beta_0 + \beta_1 (t - (k-1)), \quad (21)$$

$\hat{s}^{(2)}$ being the running average of $\hat{s}^{(1)} = \hat{s}$. Consequently $2\hat{s}_T^{(1)} - \hat{s}_T^{(2)}$ is unbiased for the trend value at T and $b_{1,T} = 2(\hat{s}_T^{(1)} - \hat{s}_T^{(2)})/(k-1)$ is unbiased for β_1 .

Proof. $\mathbf{E}[\hat{s}_t] = k^{-1} \sum_{j < k} \{\beta_0 + \beta_1(t-j)\} = \beta_0 + \beta_1 t - \beta_1 k^{-1} \sum_{j < k} j$, and $\sum_{j < k} j = k(k-1)/2$. Applying the same operator again shifts the argument by another $(k-1)/2$. The two combinations follow by solving the pair of linear equations: $2\mathbf{E}[\hat{s}^{(1)}] - \mathbf{E}[\hat{s}^{(2)}] = \beta_0 + \beta_1 t$ and $\mathbf{E}[\hat{s}^{(1)}] - \mathbf{E}[\hat{s}^{(2)}] = \beta_1(k-1)/2$. $\square$

Remark. A running average of an increasing series is systematically too low, by half the window times the slope. Reading the smoothed curve as an estimate of the current level is therefore wrong whenever there is a trend, and double smoothing exists to correct exactly that.

3.2 Exponential smoothing

Definition 10.30 (Exponential smoothing). For $w \in (0, 1)$ and a starting value $\hat{s}_0$,

$$\hat{s}_t = (1 - w)y_t + w \hat{s}_{t-1}, \quad (22)$$

and the forecast is $\hat{y}_{T+l} = \hat{s}_T$ for every $l \geq 1$. The smoothing parameter is chosen to minimise the sum of squared one-step errors $SS(w) = \sum_{t=2}^T (y_t - \hat{s}_{t-1})^2$.

Proposition 10.31 (Unrolled form and discounted least squares). (i)

$$\hat{s}_T = (1 - w) \sum_{j=0}^{T-1} w^j y_{T-j} + w^T \hat{s}_0.$$

Thus the weights on the observations sum to $1 - w^T$ and decay geometrically.

(ii) The weighted least squares estimate of β_0 in $y_t = \beta_0 + \varepsilon_t$ with weights $w_t = w^{T-t}$ is $\sum_{t \leq T} w^{T-t} y_t / \sum_{t \leq T} w^{T-t}$, which is $\hat{s}_T$ up to the starting value and the normalisation $1 - w^T$.

(iii) Under white noise the estimate in (ii) has variance tending to $\sigma^2(1-w)/(1+w)$ as $T \rightarrow \infty$, so smaller w tracks the data and larger w smooths it.

Proof. (i) Substitute (22) into itself T times. (ii) Differentiating $\sum_t w^{T-t}(y_t - b_0)^2$ in b_0 gives the stated ratio; the denominator is $(1-w^T)/(1-w)$, so the ratio is $(1-w) \sum_j w^j y_{T-j} / (1-w^T)$. (iii) The variance is $\sigma^2 \sum_t w^{2(T-t)} / (\sum_t w^{T-t})^2 \rightarrow \sigma^2(1-w)^2 / (1-w^2)$, which is $\sigma^2(1-w)/(1+w)$. $\square$

Theorem 10.32 (The model under which exponential smoothing is optimal). Let $\{y_t\}$ satisfy $\Delta y_t = \varepsilon_t - \theta \varepsilon_{t-1}$, the $ARIMA(0, 1, 1)$ model, with $|\theta| < 1$. Then the minimum mean squared error one-step forecast satisfies the exponential smoothing recursion with $w = \theta$:

$$\hat{s}_T := \mathbf{E}[y_{T+1} \mid \mathcal{L}_T] = (1 - \theta)y_T + \theta \hat{s}_{T-1}. \quad (23)$$

Proof. Taking conditional expectations in $y_{T+1} = y_T + \varepsilon_{T+1} - \theta \varepsilon_T$ and using $\mathbf{E}[\varepsilon_{T+1} \mid \mathcal{L}_T] = 0$ gives $\hat{s}_T = y_T - \theta \varepsilon_T$. Applied one step earlier the same identity reads $\hat{s}_{T-1} = y_{T-1} - \theta \varepsilon_{T-1}$, so

the one-step error is

$$y_T - \hat{s}_{T-1} = y_T - y_{T-1} + \theta \varepsilon_{T-1} = \varepsilon_T - \theta \varepsilon_{T-1} + \theta \varepsilon_{T-1} = \varepsilon_T.$$

Substituting $\varepsilon_T = y_T - \hat{s}_{T-1}$ into $\hat{s}_T = y_T - \theta \varepsilon_T$ gives $\hat{s}_T = y_T - \theta(y_T - \hat{s}_{T-1})$, which is (23). $\square$

Remark. Exponential smoothing is therefore not a heuristic: it is the optimal forecast of a nonstationary process whose differences are $MA(1)$, and the smoothing parameter is that model's moving average coefficient. The same calculation explains the flat forecast function: $\mathbf{E}[y_{T+l} \mid \mathcal{L}_T] = \hat{s}_T$ for every $l \geq 1$, since the extra innovations all have conditional mean zero.

Definition 10.33 (Holt–Winters). With smoothing parameters w_1, w_2, w_3 and seasonal base s , the level, slope and seasonal estimates are updated by

$$b_{0,t} = (1 - w_1)(y_t - \hat{S}_{t-s}) + w_1(b_{0,t-1} + b_{1,t-1}), \quad (24)$$

$$b_{1,t} = (1 - w_2)(b_{0,t} - b_{0,t-1}) + w_2 b_{1,t-1}, \quad (25)$$

$$\hat{S}_t = (1 - w_3)(y_t - b_{0,t}) + w_3 \hat{S}_{t-s}, \quad (26)$$

and the forecast is $\hat{y}_{T+l} = b_{0,T} + b_{1,T}l + \hat{S}_{T+l-s\lceil l/s \rceil}$.

Proposition 10.34 (Holt–Winters reduces to what it generalises). Dropping (26) and setting $\hat{S} \equiv 0$ leaves Holt’s method for a local linear trend; setting in addition $b_{1,t} \equiv 0$ leaves (22) with $w = w_1$.

Proof. Immediate substitution. □

3.3 Seasonal models

Definition 10.35 (Seasonal autoregression). With seasonal base s , the $SAR(P)$ model is

$$y_t = \beta_0 + \beta_1 y_{t-s} + \beta_2 y_{t-2s} + \cdots + \beta_P y_{t-Ps} + \varepsilon_t, \quad (27)$$

which omits the intervening lags that an $AR(Ps)$ would include.

Proposition 10.36 ($SAR(1)$ is s interleaved $AR(1)$ chains). Let $|\beta_1| < 1$ in the $SAR(1)$ model $y_t = \beta_0 + \beta_1 y_{t-s} + \varepsilon_t$. Then each of the s subseries $\{y_{ms+i}\}_m$, $i = 1, \dots, s$, is a stationary $AR(1)$

with coefficient β_1 , the s subseries are independent of one another, and the autocorrelation function of $\{y_t\}$ is

$$\rho_k = \begin{cases} \beta_1^{k/s}, & s \mid k, \\ 0, & \text{otherwise.} \end{cases} \quad (28)$$

Proof. Reading (27) along a fixed residue class i modulo s gives exactly Definition 10.2 with the innovations $\{\varepsilon_{ms+i}\}_m$, which are independent and identically distributed; different residue classes use disjoint sets of innovations, hence are independent. Theorem 10.3 applied within a class gives $\rho_{ms} = \beta_1^m$, and two observations in different classes are independent, so their correlation is 0. $\square$

Remark. The correlogram of a $SAR(1)$ has spikes at $s, 2s, \dots$ and nothing between, which is what distinguishes it from an $AR(s)$: the latter fills in the intervening lags. A monthly series whose correlogram is large at 12 and 24 and negligible elsewhere is asking for the seasonal model, not the twelfth-order one, which would cost eleven extra parameters.

3.4 Unit roots

Definition 10.37 (Dickey–Fuller regression). Least squares is applied to

$$\Delta y_t = \beta_0 + (\phi - 1) y_{t-1} + \beta_1 t + \varepsilon_t, \quad (29)$$

and t_{DF} is the t ratio on y_{t-1} . The hypothesis $\phi = 1$ is a random walk and $\phi < 1$ a stationary alternative, so the test is one-sided to the left. The **augmented** version adds $\sum_{j \leq p} c_j \Delta y_{t-j}$ to the right of (29) to absorb serial correlation in the errors.

Theorem 10.38 (The rate and the shape both change at the boundary). Let $y_t = y_{t-1} + \varepsilon_t$ with $y_0 = 0$ and $\mathbf{E}[\varepsilon_t^4] < \infty$, and fit y_t on y_{t-1} without intercept, giving $\hat{\phi} = \sum_t y_{t-1} y_t / \sum_t y_{t-1}^2$. Then

$$\frac{1}{T} \sum_{t=1}^T y_{t-1} \varepsilon_t = \frac{1}{2} \left(\frac{y_T^2}{T} - \frac{1}{T} \sum_{t=1}^T \varepsilon_t^2 \right) \xrightarrow{d} \frac{\sigma^2}{2} (\chi_1^2 - 1), \quad (30)$$

and

$$\mathbb{E}\left[\frac{1}{T^2} \sum_{t=1}^T y_{t-1}^2\right] = \frac{\sigma^2(T-1)}{2T} \longrightarrow \frac{\sigma^2}{2}. \quad (31)$$

So $T(\hat{\phi} - 1)$, not $\sqrt{T}(\hat{\phi} - 1)$, is the quantity of order one, and its numerator has a limit law that is not normal: it is skewed, with median at $\sigma^2(\chi_{1,0.5}^2 - 1)/2 = -0.273\sigma^2$ and no mass below $-\sigma^2/2$.

Proof. From $y_t = y_{t-1} + \varepsilon_t$, squaring gives $y_t^2 = y_{t-1}^2 + 2y_{t-1}\varepsilon_t + \varepsilon_t^2$; summing over $t \leq T$ and telescoping gives $y_T^2 = 2 \sum_t y_{t-1}\varepsilon_t + \sum_t \varepsilon_t^2$, which is the first equality in (30). Now $y_T/\sqrt{T} = T^{-1/2} \sum_t \varepsilon_t \xrightarrow{d} N(0, \sigma^2)$ by the prerequisites' central limit theorem, so $y_T^2/T \xrightarrow{d} \sigma^2 \chi_1^2$ by the continuous mapping theorem; and $T^{-1} \sum_t \varepsilon_t^2 \xrightarrow{p} \sigma^2$ by the weak law, the fourth moment making $\{\varepsilon_t^2\}$ square integrable. Slutsky's theorem combines the two.

For (31), $\mathbb{E}[y_{t-1}^2] = (t-1)\sigma^2$ and $\sum_{t=1}^T (t-1) = T(T-1)/2$. □

Remark. The exact limit law of t_{DF} is **not** derived here. Obtaining it requires the joint limit of the numerator and the denominator, and the denominator's limit is a functional of a Brownian

path, which the prerequisites do not construct. What Theorem 10.38 establishes is the two facts that matter for using the test: the estimator converges at rate T rather than $\sqrt{T}$, so the usual standard error is the wrong scale, and the limit is not normal, so normal critical values are wrong. §5.5 obtains the critical values by simulation and compares them with the ones the package uses.

Remark. The test is one-sided and its critical values lie far into the left tail: at $T = 250$ the 5% point is near -2.89 with an intercept and -3.45 with an intercept and trend, against the normal -1.64 . Using the normal value would reject a true unit root far too often. The direction of the error is the opposite of Unit 9's spurious regression, where the same nonstationarity made t ratios too large.

3.5 Volatility

Definition 10.39 (ARCH and GARCH). Let $\{z_t\}$ be independent standard normal. The $GARCH(1, 1)$ model is

$$\varepsilon_t = \sigma_t z_t, \quad \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \delta \sigma_{t-1}^2, \quad (32)$$

with $\omega > 0$, $\alpha, \delta \geq 0$ and $\alpha + \delta < 1$. The case $\delta = 0$ is $ARCH(1)$. FREES writes w, γ_1, δ_1 for

ω, α, δ .

Theorem 10.40 (A GARCH series is white noise with a non-constant conditional variance). Suppose $\{\varepsilon_t\}$ solves (32) and is weakly stationary with $\mathbf{E}[\varepsilon_t^2] < \infty$. Then

$$\mathbf{E}[\varepsilon_t] = 0, \quad \text{cov}(\varepsilon_t, \varepsilon_{t+k}) = 0 \ (k \neq 0), \quad \text{var}(\varepsilon_t) = \frac{\omega}{1 - \alpha - \delta}, \quad (33)$$

while the conditional variance given the past is σ_t^2 , which depends on t . Moreover $\{\varepsilon_t^2\}$ satisfies

$$\varepsilon_t^2 = \omega + (\alpha + \delta)\varepsilon_{t-1}^2 + \nu_t - \delta\nu_{t-1}, \quad \nu_t = \varepsilon_t^2 - \sigma_t^2, \quad (34)$$

an $ARMA(1, 1)$ whose innovations ν_t are uncorrelated with mean zero.

Proof. σ_t is determined by $\varepsilon_{t-1}, \varepsilon_{t-2}, \dots$, hence is independent of z_t , so $\mathbf{E}[\varepsilon_t] = \mathbf{E}[\sigma_t] \mathbf{E}[z_t] = 0$ and, for $k > 0$, $\mathbf{E}[\varepsilon_t \varepsilon_{t+k}] = \mathbf{E}[\varepsilon_t \sigma_{t+k}] \mathbf{E}[z_{t+k}] = 0$, $\varepsilon_t \sigma_{t+k}$ being determined by innovations dated before $t + k$. Taking expectations in (32), $\mathbf{E}[\varepsilon_t^2] = \mathbf{E}[\sigma_t^2]$ because $\mathbf{E}[z_t^2] = 1$, so weak stationarity gives $\mathbf{E}[\varepsilon^2] = \omega + \alpha \mathbf{E}[\varepsilon^2] + \delta \mathbf{E}[\varepsilon^2]$ and hence (33).

For (34), substitute $\sigma_t^2 = \varepsilon_t^2 - \nu_t$ and $\sigma_{t-1}^2 = \varepsilon_{t-1}^2 - \nu_{t-1}$ into (32):

$$\varepsilon_t^2 - \nu_t = \omega + \alpha \varepsilon_{t-1}^2 + \delta(\varepsilon_{t-1}^2 - \nu_{t-1}),$$

which rearranges to the stated form. That $\nu_t = \sigma_t^2(z_t^2 - 1)$ has mean zero and is uncorrelated across t follows as above, $\mathbf{E}[z_t^2 - 1] = 0$ and σ_t^2 being determined by the past. $\square$

Theorem 10.41 (Conditional normality produces unconditional fat tails). For a weakly stationary $ARCH(1)$ with $3\alpha^2 < 1$,

$$\mathbf{E}[\varepsilon_t^4] = \frac{3\omega^2(1 + \alpha)}{(1 - \alpha)(1 - 3\alpha^2)}, \quad \frac{\mathbf{E}[\varepsilon_t^4]}{\{\mathbf{E}[\varepsilon_t^2]\}^2} = \frac{3(1 - \alpha^2)}{1 - 3\alpha^2} > 3 \quad \text{for } \alpha > 0. \quad (35)$$

Proof. $\mathbf{E}[\varepsilon_t^4] = \mathbf{E}[\sigma_t^4 z_t^4] = 3 \mathbf{E}[\sigma_t^4]$, the fourth moment of a standard normal being 3 and σ_t independent of z_t . With $\delta = 0$,

$$\mathbf{E}[\sigma_t^4] = \mathbf{E}[(\omega + \alpha \varepsilon_{t-1}^2)^2] = \omega^2 + 2\omega\alpha \mathbf{E}[\varepsilon^2] + \alpha^2 \mathbf{E}[\varepsilon^4],$$

and $\mathbf{E}[\varepsilon^2] = \omega/(1 - \alpha)$ by (33). Substituting and solving for $\mathbf{E}[\varepsilon^4]$,

$$\mathbf{E}[\varepsilon^4](1 - 3\alpha^2) = 3\omega^2 \left(1 + \frac{2\alpha}{1 - \alpha}\right) = \frac{3\omega^2(1 + \alpha)}{1 - \alpha}.$$

Dividing by $\{\omega/(1 - \alpha)\}^2$ gives the kurtosis $3(1 + \alpha)(1 - \alpha)/(1 - 3\alpha^2)$, and $1 - \alpha^2 > 1 - 3\alpha^2 > 0$ for $\alpha > 0$. $\square$

Remark. The innovations are conditionally normal at every date, and the series is nonetheless leptokurtic: at $\alpha = 0.5$ the kurtosis is 9. Mixing normals of different variances produces heavy tails, which is what a plot of daily returns shows.

Definition 10.42 (Conditional likelihood). Given $\varepsilon_1, \dots, \varepsilon_n$ and a starting value σ_1^2 , the recursion (32) determines $\sigma_2^2, \dots, \sigma_n^2$, and the parameters are estimated by maximising

$$\ell(\omega, \alpha, \delta) = -\frac{1}{2} \sum_{t=2}^n \left\{ \ln(2\pi\sigma_t^2) + \frac{\varepsilon_t^2}{\sigma_t^2} \right\}. \quad (36)$$

Remark. The likelihood (36) is not concave in (ω, α, δ) , so the maximum likelihood theorem of the prerequisites does not apply and the standard errors it would supply are not available here. §5.6 maximises (36) directly and checks the estimates against a package, which is what can be verified; the asymptotic theory of GARCH estimation is not developed in this course.

4 Regression Trees

ISLR §8.1.1.

4.1 The predictor

Definition 10.43 (Partition predictor). Let $R_1, \dots, R_J$ be disjoint sets covering the predictor space. The associated **partition predictor** is

$$\hat{f}(\mathbf{x}) = \sum_{j=1}^J \hat{c}_j \mathbf{1}\{\mathbf{x} \in R_j\}, \quad (37)$$

constant on each region. A **regression tree** is a partition predictor whose regions are boxes obtained by successive splits on one coordinate at a time.

Proposition 10.44 (The region constants are the region means). For fixed regions, the values $\hat{c}_1, \dots, \hat{c}_J$ minimising $\sum_{i=1}^n \{y_i - \hat{f}(\mathbf{x}_i)\}^2$ are the region means $\hat{c}_j = n_j^{-1} \sum_{i:\mathbf{x}_i \in R_j} y_i$, where $n_j = |\{i : \mathbf{x}_i \in R_j\}|$, and the minimised value is $\sum_j \sum_{i \in R_j} (y_i - \hat{c}_j)^2$.

Proof. The criterion separates into J sums, the j th involving only c_j ; each is minimised at the mean of its observations (Unit 2 with a single intercept). $\square$

Proposition 10.45 (A tree is a linear model on indicators). Let $\mathbf{D}$ be the $n \times J$ matrix with $D_{ij} = \mathbb{1}\{\mathbf{x}_i \in R_j\}$. For fixed regions the fitted values of (37) are $\mathbf{D}(\mathbf{D}'\mathbf{D})^{-1}\mathbf{D}'\mathbf{Y}$, the orthogonal projection of $\mathbf{Y}$ on the column space of $\mathbf{D}$, and $\mathbf{D}'\mathbf{D} = \text{diag}(n_1, \dots, n_J)$.

Proof. The regions are disjoint and cover the space, so each row of $\mathbf{D}$ has exactly one 1 and the columns are orthogonal with squared norms n_j . Least squares on $\mathbf{D}$ is Proposition 10.44, and Unit 2 identifies the fitted values with the projection. $\square$

Remark. The columns of $\mathbf{D}$ are the seasonal dummies of Unit 9 with the calendar replaced by a partition of the predictor space; the identifiability discussion there applies verbatim, which is why (37) carries no separate intercept. What makes a tree different from an ordinary dummy regression is not the fit given the regions, which is a projection, but that the regions are chosen from the same data.

4.2 Recursive binary splitting

Remark. Minimising the residual sum of squares over all partitions into J boxes is not a computation that can be carried out. Even the restricted family in which each of p predictors is cut at one of $n - 1$ places, applied recursively, has more members than can be enumerated: the first split alone admits $p(n - 1)$ choices, and each subsequent split multiplies the count again. Trees are therefore grown greedily, one split at a time.

Definition 10.46 (Split). A split of a node R on coordinate j at cutpoint c produces $R_L = R \cap \{\mathbf{x} : x_j < c\}$ and $R_R = R \cap \{\mathbf{x} : x_j \geq c\}$. Recursive binary splitting chooses, at each node, the (j, c) minimising

$$\sum_{i \in R_L} (y_i - \bar{y}_L)^2 + \sum_{i \in R_R} (y_i - \bar{y}_R)^2, \quad (38)$$

and stops when a node has fewer than a preset number of observations.

Theorem 10.47 (What a split gains). Let a node contain n observations with mean $\bar{y}$, and let a

split send n_L of them to the left with mean $\bar{y}_L$ and $n_R = n - n_L$ to the right with mean $\bar{y}_R$. Then

$$\sum_i (y_i - \bar{y})^2 = \sum_{i \in R_L} (y_i - \bar{y}_L)^2 + \sum_{i \in R_R} (y_i - \bar{y}_R)^2 + \frac{n_L n_R}{n} (\bar{y}_L - \bar{y}_R)^2. \quad (39)$$

Minimising (38) is therefore the same as maximising $n_L n_R (\bar{y}_L - \bar{y}_R)^2 / n$, and the gain is zero exactly when the two region means coincide.

Proof. Split the left sum at $\bar{y}_L$: $\sum_{i \in R_L} (y_i - \bar{y})^2 = \sum_{i \in R_L} (y_i - \bar{y}_L)^2 + n_L (\bar{y}_L - \bar{y})^2$, the cross term vanishing because $\sum_{i \in R_L} (y_i - \bar{y}_L) = 0$; similarly on the right. Adding,

$$\sum_i (y_i - \bar{y})^2 = \sum_{i \in R_L} (y_i - \bar{y}_L)^2 + \sum_{i \in R_R} (y_i - \bar{y}_R)^2 + n_L (\bar{y}_L - \bar{y})^2 + n_R (\bar{y}_R - \bar{y})^2.$$

Since $n\bar{y} = n_L \bar{y}_L + n_R \bar{y}_R$, one has $\bar{y}_L - \bar{y} = (n_R/n)(\bar{y}_L - \bar{y}_R)$ and $\bar{y}_R - \bar{y} = -(n_L/n)(\bar{y}_L - \bar{y}_R)$, so the last two terms are $\{n_L n_R^2 + n_R n_L^2\}(\bar{y}_L - \bar{y}_R)^2 / n^2 = n_L n_R (\bar{y}_L - \bar{y}_R)^2 / n$. $\square$

Corollary 10.48 (Cost of one node). Sorting each predictor once, the best split of a node of size n over p predictors is found in $O(pn)$ further operations.

Proof. Sweeping the sorted values of one predictor from left to right, n_L increases by one at a time and $\sum_{i \in R_L} y_i$ is maintained by a single addition, so $\bar{y}_L$, $\bar{y}_R$ and the gain of Theorem 10.47 are each computed in a fixed number of operations per candidate cutpoint. There are $n - 1$ cutpoints per predictor. $\square$

Remark. The identity (39) is the one-way analysis of variance decomposition of Unit 3 with two groups. It says a split is judged by the separation of the two group means weighted by $n_L n_R / n$, which penalises splits that peel off a handful of observations however extreme they are.

Proposition 10.49 (Monotone invariance). Replacing a predictor x_j by $g(x_j)$ for any strictly increasing g leaves every tree grown by Definition 10.46 unchanged, up to relabelling the cutpoints.

Proof. The candidate splits of coordinate j are determined by the ordering of the observed values, which g preserves, and the criterion (38) depends on the split only through which observations fall

on each side. □

Remark. A tree therefore needs no decision about logarithms, standardisation or outlying predictor values, all of which change a linear fit. It pays for that with the two weaknesses of a piecewise-constant predictor: the fitted surface is a step function, so a genuinely linear relation is approximated by a staircase that needs many splits, and the prediction at any $\mathbf{x}$ outside the range of the training data equals the prediction at the nearest corner of the box, so a tree cannot extrapolate at all. §6.4 exhibits both.

Remark. Nothing above says when to stop. Growing until each leaf is small overfits, and the repair, cost-complexity pruning with the penalty chosen by cross-validation, is Unit 11.

5 Computation: Formulas Against Package Output

FREES §8.3, §8.6, §9.2, §9.4–9.5; ISLR §8.3.2.

Three series and one cross-sectional data set. **LakeHuron**: annual mean level of Lake Huron in feet, $T = 98$, a textbook stationary autoregression. **EuStockMarkets**: the DAX column again, whose log differences carry no autocorrelation in the level and a great deal in the squares. **Nile**: annual flow of

the Nile at Aswan, $T = 100$. `Boston` from `ISLR2` for the tree.

```
1 y <- as.numeric(LakeHuron); T <- length(y)
2 dl <- diff(log(as.numeric(EuStockMarkets[, "DAX"])))
3 c(T_huron = T, mean = mean(y), sd = sd(y),
4   n_dax = length(dl), sd_dax = sd(dl))
```

```
##      T_huron      mean      sd      n_dax      sd_dax
## 98.0000 579.0041 1.3183 1859.0000 0.0103
```

5.1 Identification (§1.1, §1.3)

Theorem 10.3 predicts $\rho_k = \beta_1^k$ and Theorem 10.12 predicts $\alpha_{k,k} = 0$ beyond the order. Both are read off the same series.

```
1 r <- acf(y, plot = FALSE, lag.max = 5)$acf[2:6]
2 a <- pacf(y, plot = FALSE, lag.max = 5)$acf[1:5]
3 round(rbind(acf = r, geometric_prediction = r[1]^(1:5), pacf = a), 4)
```

```
##           [,1]      [,2]      [,3]      [,4]      [,5]
## acf          0.8319  0.6099  0.4583  0.3705  0.3256
## geometric_prediction 0.8319  0.6921  0.5757  0.4790  0.3985
## pacf          0.8319 -0.2668  0.1308  0.0341  0.0621
```

The correlogram decays but faster than r_1^k at every lag, and the partial correlogram is not negligible at lag 2, against a band of $\pm 2/\sqrt{98} = \pm 0.202$: an $AR(2)$ is indicated, not an $AR(1)$. Solving the Yule–Walker system (10) with the sample autocorrelations reproduces **pacf** exactly; the regression version does not.

```

1 lagm <- function(v, k) c(rep(NA, k), head(v, -k))
2 rk <- acf(y, plot = FALSE, lag.max = 6)$acf[-1]
3 yw <- sapply(1:5, function(k) {
4   R <- outer(1:k, 1:k, function(i, j) c(1, rk)[abs(i - j) + 1])
5   solve(R, rk[1:k])[k] })
6 lsq <- sapply(1:5, function(k) {
7   Z <- sapply(1:k, function(j) lagm(y, j)); coef(lm(y ~ Z))[k + 1] })
8 round(rbind(package = a, yule_walker = yw, package_minus_yw = a - yw,
9             least_squares = unname(lsq)), 6)

```

##	[,1]	[,2]	[,3]	[,4]	[,5]
## package	0.8319	-0.2668	0.1308	0.03406	0.06209
## yule_walker	0.8319	-0.2668	0.1308	0.03406	0.06209
## package_minus_yw	0.0000	0.0000	0.0000	0.00000	0.00000
## least_squares	0.8364	-0.2376	0.1088	0.06249	0.02561

The end effect is $O(T^{-1})$: multiplying the gap by T leaves a constant.

```

1 set.seed(19)
2 t(sapply(c(50, 200, 800), function(n) {
3   g <- replicate(400, {
4     z <- as.numeric(arima.sim(list(ar = c(0.6, -0.3)), n))
5     q <- acf(z, plot = FALSE, lag.max = 2)$acf[-1]
6     R <- outer(1:2, 1:2, function(i, j) c(1, q)[abs(i - j) + 1])
7     solve(R, q)[2] - coef(lm(z ~ cbind(lagm(z, 1), lagm(z, 2))))[3] })
8   c(T = n, mean_absolute_gap = mean(abs(g)), gap_times_T = mean(abs(g))*n) })

```

```

##           T mean_absolute_gap gap_times_T
## [1,]  50           0.018383      0.9192
## [2,] 200           0.004388      0.8776
## [3,] 800           0.001089      0.8713

```

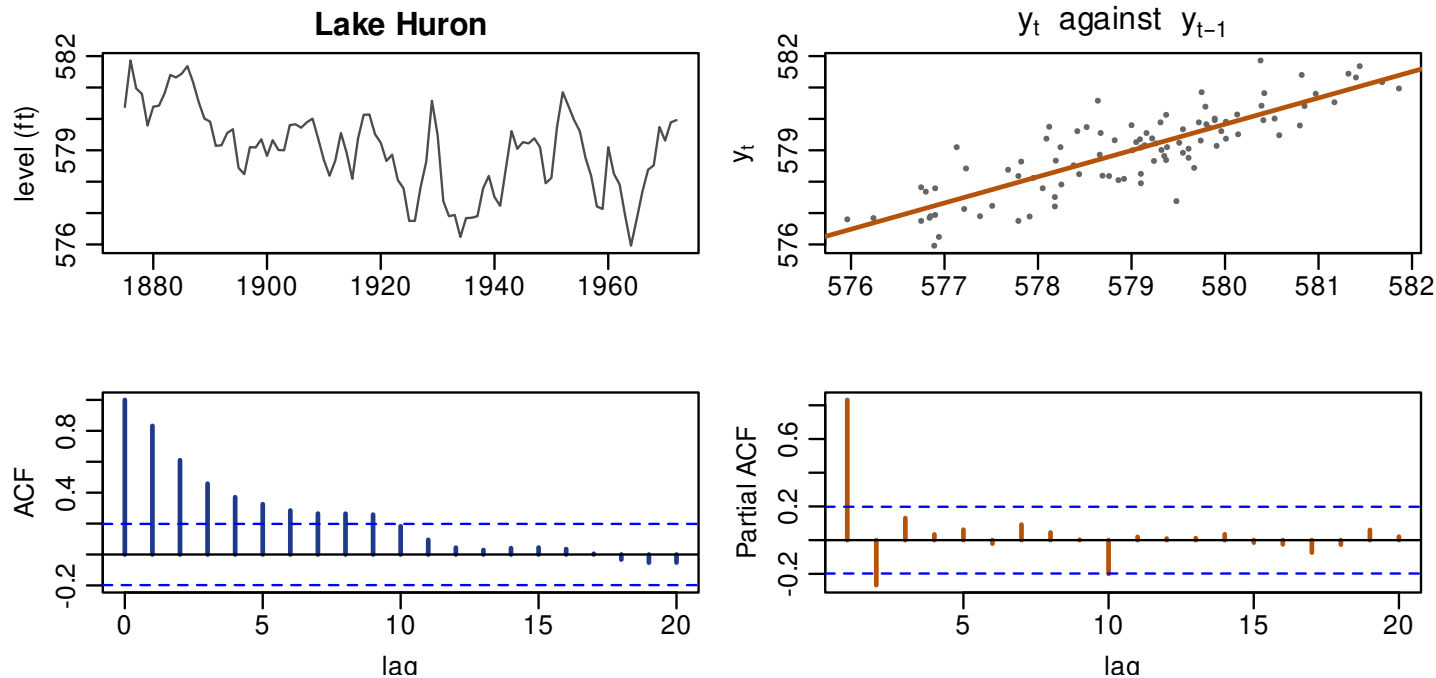


Figure 10.1: The series and the three pictures that identify its model. The scatter of y_t against y_{t-1} has slope $\hat{\beta}_1$; the correlogram decays geometrically as Theorem 10.3 requires of an autoregression, unlike the near-flat correlogram of a random walk; the partial correlogram is large at lag 1, appreciable at lag 2 and negligible thereafter.

5.2 Conditional least squares and its asymptotics (§2.1–§2.2)

Proposition 10.15 four ways: the closed form, `lm`, `arima` with the conditional sum of squares method, and r_1 .

```

1 m1 <- lm(y[-1] ~ y[-T]); b1 <- unname(coef(m1)[2]); e <- residuals(m1)
2 num <- sum((y[-T] - mean(y[-T])) * (y[-1] - mean(y[-1])))
3 den <- sum((y[-T] - mean(y[-T]))^2)
4 c(b1_closed_form = num/den, b1_lm = b1,
5   b1_arima_css = unname(coef(arima(y, order = c(1,0,0), method = "CSS"))[1]),
6   r1 = acf(y, plot = FALSE)$acf[2],
7   s_lm = summary(m1)$sigma,
8   s_formula = sqrt(sum((e - mean(e))^2)/(T - 3)))

```

##	b1_closed_form	b1_lm	b1_arima_css	r1
##	0.8364	0.8364	0.8364	0.8319
##	s_lm	s_formula		
##	0.7209	0.7209		

Theorem 10.16 says $\sqrt{T}(\hat{\beta}_1 - \beta_1) \xrightarrow{d} N(0, 1 - \beta_1^2)$, and Corollary 10.17 says the regression standard error already estimates that.

```

1 c(se_reported = summary(m1)$coef[2, 2],
2   se_asymptotic = sqrt((1 - b1^2)/(T - 1)),
3   difference = summary(m1)$coef[2, 2] - sqrt((1 - b1^2)/(T - 1)),
4   implied_gamma0 = summary(m1)$sigma^2/(1 - b1^2),
5   sample_variance = var(y))

```

```

##      se_reported      se_asymptotic      difference
##      0.05567899      0.05565136      0.00002764
## implied_gamma0 sample_variance
##      1.73011062      1.73791100

```

The two standard errors differ in the fifth decimal. The estimator is nonetheless biased downwards, and the bias is $O(T^{-1})$ against a standard deviation of order $T^{-1/2}$.

```

1 set.seed(31); B <- 4000
2 t(sapply(c(50, 100, 400), function(n) {
3   b <- replicate(B, { z <- as.numeric(arima.sim(list(ar = 0.8), n))
4                     coef(lm(z[-1] ~ z[-n]))[2] })
5   c(T = n, mean_b1 = mean(b), bias = mean(b) - 0.8, sd_b1 = sd(b),
6     asymptotic_sd = sqrt((1 - 0.8^2)/n),

```

```
7 bias_over_sd = (mean(b) - 0.8)/sd(b)) }))
```

```
##          T mean_b1      bias    sd_b1 asymptotic_sd bias_over_sd
## [1,]   50  0.7291 -0.070918 0.10483      0.08485      -0.6765
## [2,]  100  0.7649 -0.035075 0.06685      0.06000      -0.5247
## [3,]  400  0.7908 -0.009171 0.03072      0.03000      -0.2985
```

The bias roughly halves when T doubles while the standard deviation falls by $\sqrt{2}$, so the ratio of the two shrinks and the bias becomes irrelevant.

5.3 Correlograms and portmanteau tests (§2.3)

Theorem 10.18 gives $\sqrt{T} r_k \xrightarrow{d} N(0, 1)$, so Corollary 10.19 makes the $\pm 2/\sqrt{T}$ band a 95.45% band. Unit 9 measured 95.6% and 95.8% without being able to prove it.

```
1 set.seed(3); B <- 5000
2 t(sapply(c(50, 200), function(n) {
3   rk <- replicate(B, acf(rnorm(n), plot = FALSE, lag.max = 1)$acf[2])
4   c(T = n, sd_r1 = sd(rk), one_over_sqrt_T = 1/sqrt(n),
5     inside_band = mean(abs(rk) < 2/sqrt(n)), limit = 1 - 2*pnorm(-2)) }))
```

```
##          T    sd_r1 one_over_sqrt_T inside_band  limit
```

```
## [1,] 50 0.14015      0.14142      0.9556 0.9545
## [2,] 200 0.07037     0.07071      0.9576 0.9545
```

Theorem 10.21 gives $Q_{LB} \xrightarrow{d} \chi_K^2$ for an unfitted series. The degrees-of-freedom reduction for residuals of a fitted model was stated but not proved; it is measured here.

```
1 set.seed(11); B <- 3000; n <- 200; K <- 10
2 Q <- function(x) { rk <- acf(x, plot = FALSE, lag.max = K)$acf[2:(K+1)]
3   m <- length(x); m*(m + 2)*sum(rk^2/(m - 1:K)) }
4 q0 <- replicate(B, Q(rnorm(n)))
5 q1 <- replicate(B, { z <- as.numeric(arima.sim(list(ar = 0.6), n))
6   Q(residuals(lm(z[-1] ~ z[-n]))) })
7 rbind(white_noise = c(mean_Q = mean(q0), reject_at_K = mean(q0 > qchisq(.95, K)),
8   reject_at_K_minus_1 = mean(q0 > qchisq(.95, K - 1))),
9   AR1_residuals = c(mean(q1), mean(q1 > qchisq(.95, K)),
10   mean(q1 > qchisq(.95, K - 1))))
```

```
##          mean_Q reject_at_K reject_at_K_minus_1
## white_noise 10.041    0.05367         0.08233
## AR1_residuals 8.878    0.02933         0.04667
```

The mean of Q_{LB} is $K = 10$ for the unfitted series and $K - 1 = 9$ for the residuals of a fitted $AR(1)$,

and only the reduced degrees of freedom give the nominal 5%. Applied to Lake Huron with $K = 10$, neither order is rejected.

```
1 m2 <- lm(y[-(1:2)] ~ y[-c(1, T)] + y[-((T-1):T)])
2 lb <- function(res, fitdf) unlist(Box.test(res, lag = 10,
3                                     type = "Ljung-Box", fitdf = fitdf)[1:3])
4 rbind(AR1 = c(lb(residuals(m1), 1), s = summary(m1)$sigma),
5       AR2 = c(lb(residuals(m2), 2), s = summary(m2)$sigma))
```

##	statistic.X-squared	parameter.df	p.value	s
## AR1	11.845	9	0.2222	0.7209
## AR2	5.205	8	0.7354	0.6846

A portmanteau statistic spread over ten lags has little power against a single missing one. The case for the second order rests instead on the partial correlogram, whose lag two value lies outside the band, on the t ratio of the second coefficient, and on the residual standard deviation.

```
1 round(summary(m2)$coef, 4)
```

##	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	124.9499	32.0626	3.897	0.0002
## y[-c(1, T)]	1.0217	0.0975	10.483	0.0000

```
## y[-((T - 1):T)] -0.2376      0.0971  -2.446   0.0163
```

5.4 Forecasting (§2.4)

The chain rule of Definition 10.22 against the closed form (16), and the error variance (17) against its limit $\gamma(0)$.

```
1 b0 <- unname(coef(m1)[1]); s <- summary(m1)$sigma; mu <- b0/(1 - b1)
2 ch <- numeric(9); ch[1] <- y[T]
3 for (k in 2:9) ch[k] <- b0 + b1*ch[k - 1]
4 l <- 1:8
5 round(rbind(chain_rule = ch[-1], closed_form = mu + b1^l*(y[T] - mu),
6             half_width = 2*s*sqrt(cumsum(b1^(2*(1 - 1))))),
7             fraction_of_gamma0 = cumsum(b1^(2*(1 - 1)))*(1 - b1^2)), 4)
```

```
##           [,1]      [,2]      [,3]      [,4]
## chain_rule 579.7977 579.6619 579.5484 579.4534
## closed_form 579.7977 579.6619 579.5484 579.4534
## half_width  1.4419   1.8797   2.1333   2.2941
## fraction_of_gamma0 0.3004   0.5106   0.6576   0.7605
##           [,5]      [,6]      [,7]      [,8]
## chain_rule 579.3739 579.3075 579.252 579.2054
```

## closed_form	579.3739	579.3075	579.252	579.2054
## half_width	2.4002	2.4717	2.521	2.5541
## fraction_of_gamma0	0.8324	0.8828	0.918	0.9426

The point forecasts converge to $\hat{\mu} = 578.968$ and the interval half-width to $2\sqrt{\hat{\gamma}(0)}$: eight years ahead the model already knows almost nothing beyond the marginal distribution of the series.

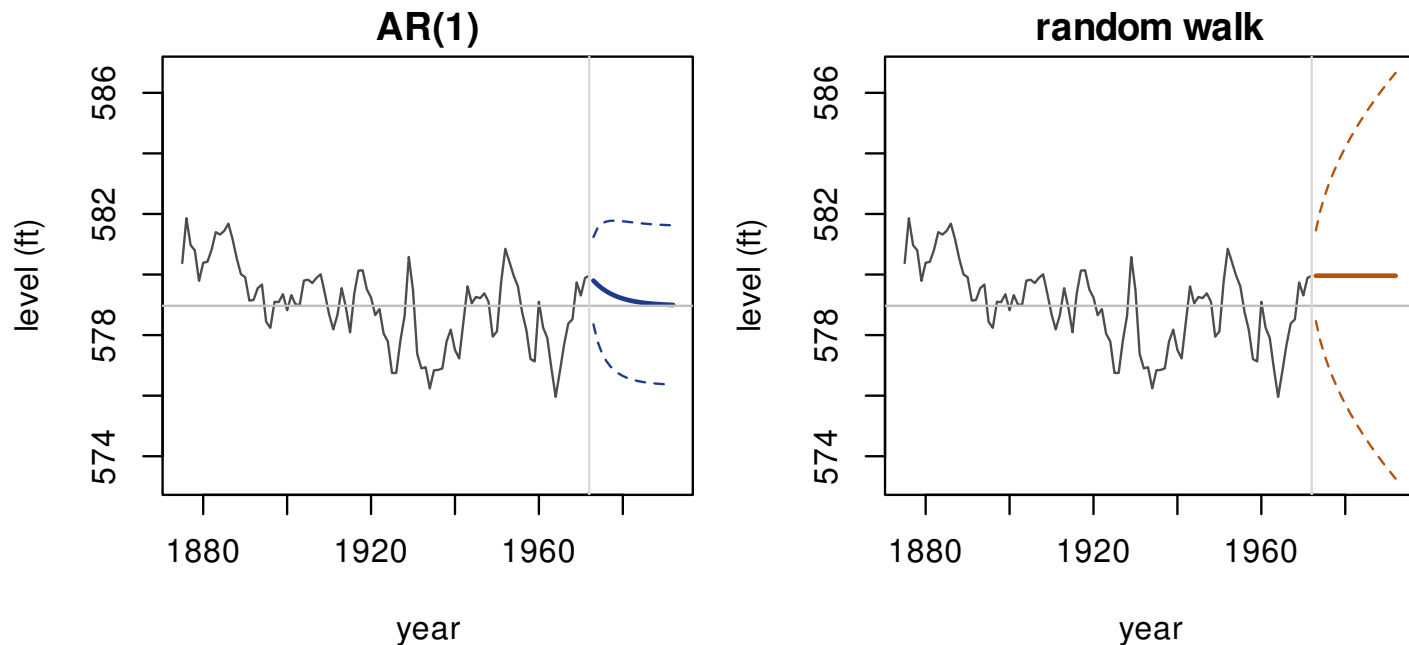


Figure 10.2: Twenty-year forecasts of the same series under the two models. The stationary $AR(1)$ reverts to $\hat{\mu}$ with an interval widening to $2\sqrt{\hat{\gamma}(0)}$ and no further (Theorem 10.23); the random walk stays at the last observation with an interval widening like $\sqrt{l}$ without bound (Unit 9). The two agree for one step and disagree completely after twenty.

5.5 Unit roots (§3.4)

The algebraic identity behind (30), and its limit law.

```
1 set.seed(3); B <- 8000; n <- 250
2 nm <- replicate(B, { u <- rnorm(n); z <- cumsum(u)
3   c(sum(z[-n]*u[-1])/n, (z[n]^2/n - mean(u^2))/2) })
4 c(identity_max_gap = max(abs(nm[1, ] - nm[2, ])),
5   mean = mean(nm[1, ]), sd = sd(nm[1, ]), sd_of_half_chisq = sqrt(1/2))
```

```
## identity_max_gap          mean          sd
##          1.776e-15        -6.145e-03      7.031e-01
## sd_of_half_chisq
##          7.071e-01
```

```
1 round(rbind(simulated = quantile(nm[1, ], c(.05, .25, .5, .75, .95)),
2         half_chisq_minus_one = (qchisq(c(.05,.25,.5,.75,.95), 1) - 1)/2), 3)
```

```
##              5%    25%    50%    75%    95%
## simulated      -0.523 -0.436 -0.266 0.136 1.411
## half_chisq_minus_one -0.498 -0.449 -0.273 0.162 1.421
```

The limit is not normal: it is skewed, its quartiles at -0.45 and 0.16 straddling a median of -0.27 ,

with a long right tail. The t statistic inherits that.

```

1 set.seed(21); B <- 8000; n <- 250
2 tdf <- function(trend) replicate(B, {
3   z <- cumsum(rnorm(n + 1)); dz <- diff(z); z1 <- z[-(n + 1)]
4   tv <- seq_along(dz)
5   if (trend) summary(lm(dz ~ z1 + tv))$coef[2, 3]
6   else       summary(lm(dz ~ z1))$coef[2, 3] })
7 round(rbind(constant_only = quantile(tdf(FALSE), c(.01, .05, .10)),
8             with_trend = quantile(tdf(TRUE), c(.01, .05, .10)),
9             package_table_with_trend = -c(3.99, 3.43, 3.13),
10            normal_reference = qnorm(c(.01, .05, .10))), 3)

```

##	1%	5%	10%
## constant_only	-3.473	-2.890	-2.574
## with_trend	-4.010	-3.449	-3.154
## package_table_with_trend	-3.990	-3.430	-3.130
## normal_reference	-2.326	-1.645	-1.282

`tseries::adf.test` fits (29) with a trend, and the third row is its own tabulated critical values at this sample size, read out of the function's table. The simulated and tabulated rows agree to about

0.02; the normal row is nowhere near either.

```

1 suppressMessages(library(tseries))
2 lev <- suppressWarnings(adf.test(as.numeric(EuStockMarkets[, "DAX"])))
3 dif <- suppressWarnings(adf.test(d1))
4 hur <- suppressWarnings(adf.test(y))
5 rbind(DAX_level = c(t_DF = unname(lev$statistic), p = lev$p.value),
6       DAX_log_difference = c(unname(dif$statistic), dif$p.value),
7       LakeHuron = c(unname(hur$statistic), hur$p.value))

```

```

##              t_DF      p
## DAX_level      -0.8207 0.9598
## DAX_log_difference -11.1046 0.0100
## LakeHuron       -2.7796 0.2540

```

The DAX is indistinguishable from a random walk and its log differences are not. Lake Huron is not rejected either, at $p = 0.25$, even though §5.2 fitted it as a stationary $AR(1)$ with $\hat{\beta}_1 = 0.84$. That is a failure of power, not evidence of a unit root, and the size of the failure can be measured: the test is run on series generated from exactly that model.

```

1 set.seed(23)
2 pv <- replicate(600, suppressWarnings(

```

```

3   adf.test(as.numeric(arima.sim(list(ar = 0.836), 98)))$p.value))
4   c(T = 98, phi = 0.836, reject_at_5_percent = mean(pv <= 0.05),
5     median_p_value = median(pv))

```

```

##              T              phi reject_at_5_percent
##          98.0000          0.8360          0.2467
## median_p_value
##          0.1757

```

Against a true $AR(1)$ with $\beta_1 = 0.84$ at $T = 98$ the test rejects a quarter of the time. Its default augmentation fits a trend and four lagged differences on 93 usable dates, and most of the information goes to them. A failure to reject is therefore weak evidence, and the correlogram and partial correlogram of §5.1 are the stronger argument here.

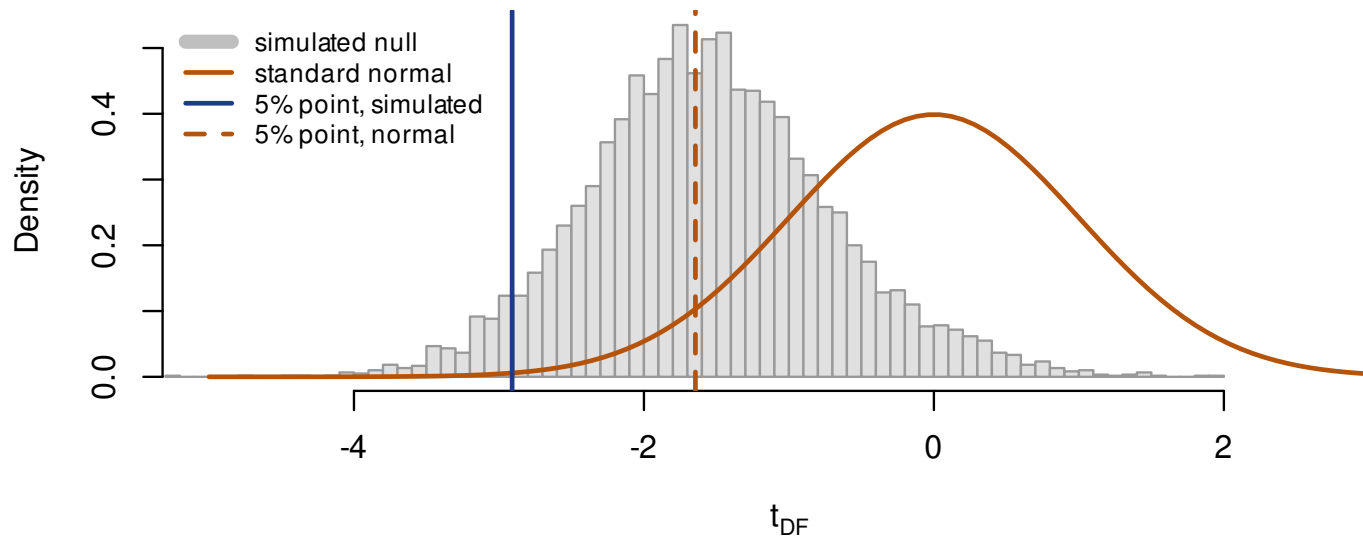


Figure 10.3: The null distribution of t_{DF} at $T = 250$ against the standard normal. It is shifted left and skewed, and its 5% point (solid) is at -2.89 where the normal's (dashed) is at -1.64 . Theorem 10.38 explains the shift: the numerator's limit is a recentred χ_1^2 , which puts most of its mass below zero.

5.6 Volatility (§3.5)

Theorem 10.40 says the returns are uncorrelated while the squares are not, and Theorem 10.41 says the kurtosis exceeds 3.

```

1 ed <- dl - mean(dl)
2 c(n = length(ed), kurtosis = mean(ed^4)/mean(ed^2)^2,
3   acf1_returns = acf(ed, plot = FALSE)$acf[2],
4   acf1_squares = acf(ed^2, plot = FALSE)$acf[2],
5   acf2_squares = acf(ed^2, plot = FALSE)$acf[3],
6   LB_returns_p = Box.test(ed, 10, "Ljung-Box")$p.value,
7   LB_squares_p = Box.test(ed^2, 10, "Ljung-Box")$p.value)

```

```

##           n      kurtosis acf1_returns acf1_squares
## 1859.0000000    9.2796890  -0.0004346    0.0787478
## acf2_squares LB_returns_p LB_squares_p
##      0.1677549    0.7836711    0.0000000

```

The conditional likelihood (36) is maximised directly and compared with **garch** from **tseries**. Both condition on the first observation.

```

1 nll <- function(p, x) {
2   w <- exp(p[1]); a1 <- plogis(p[2]); de <- plogis(p[3])*(1 - a1)

```

```

3   v <- numeric(length(x)); v[1] <- mean(x^2)
4   for (t in 2:length(x)) v[t] <- w + al*x[t-1]^2 + de*v[t-1]
5   0.5*sum(log(2*pi*v[-1]) + x[-1]^2/v[-1]) }
6   op <- optim(c(log(1e-6), qlogis(0.07), qlogis(0.95)), nll, x = ed,
7             method = "BFGS", control = list(reltol = 1e-14, maxit = 2000))
8   mn <- c(exp(op$par[1]), plogis(op$par[2]),
9         plogis(op$par[3])*(1 - plogis(op$par[2])))
10  pk <- garch(ed, order = c(1, 1), trace = FALSE)
11  round(rbind(manual = c(omega = mn[1], alpha = mn[2], delta = mn[3]),
12               tseries = unname(pk$coef),
13               difference = mn - unname(pk$coef)), 8)

```

```

##              omega      alpha      delta
## manual      0.00000476 0.0684523 0.8875734
## tseries     0.00000475 0.0683705 0.8877458
## difference  0.00000001 0.0000818 -0.0001724

```

```

1   c(loglik_manual = -op$value, loglik_tseries = c(logLik(pk)),
2     difference = -op$value - c(logLik(pk)))

```

```

## loglik_manual loglik_tseries      difference

```

```
## 5963.02767442 5963.02764358 0.00003085
```

Theorem 10.40 then gives the unconditional variance, which is compared with the sample one.

```
1 c(persistence = mn[2] + mn[3],  
2   implied_sd = sqrt(mn[1]/(1 - mn[2] - mn[3])), sample_sd = sd(ed),  
3   half_life_days = log(0.5)/log(mn[2] + mn[3]))
```

```
## persistence      implied_sd      sample_sd half_life_days  
##           0.9560           0.0104           0.0103           15.4134
```

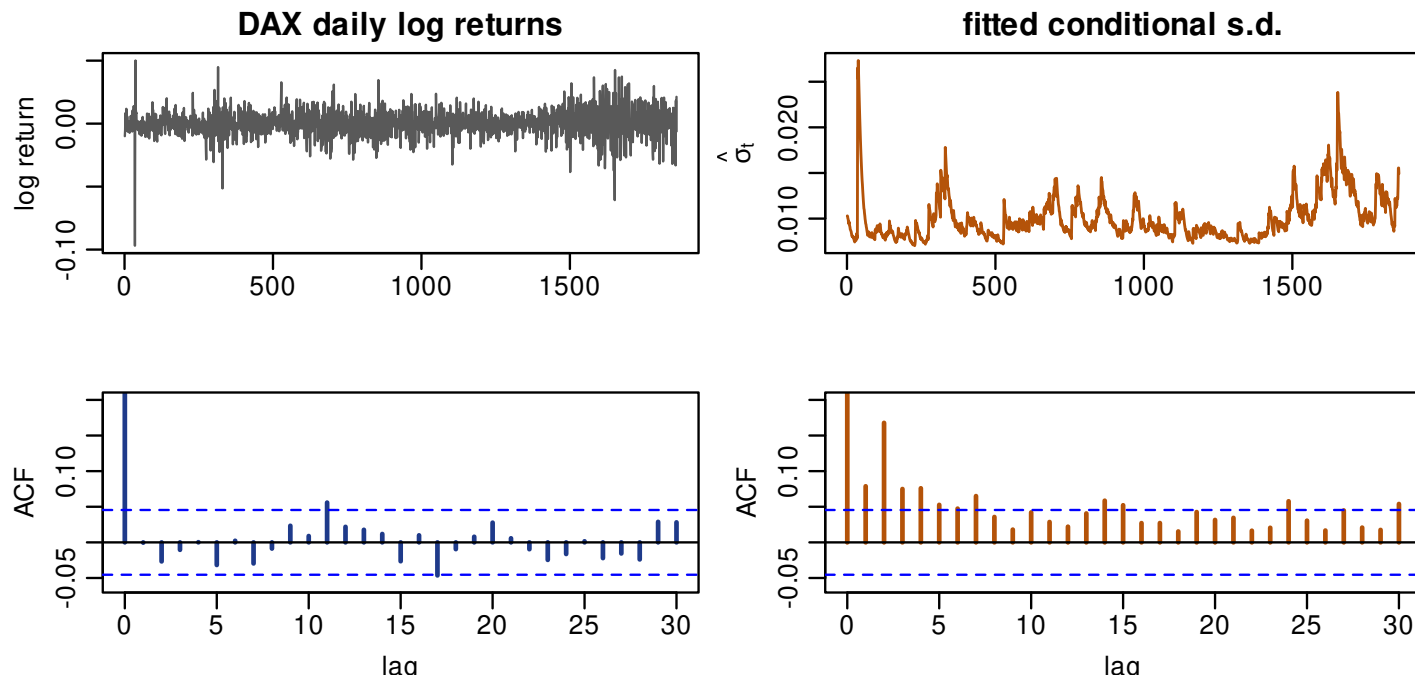


Figure 10.4: Volatility clustering. The returns pass every test for white noise in the level and fail every one in the squares, which is exactly what Theorem 10.40 describes: zero autocorrelation with a conditional variance that follows an $ARMA(1,1)$ in ε_t^2 . The vertical axes of the two correlograms are cropped at 0.20, so the spike of height 1 at lag 0 runs off the top.

5.7 Smoothing (§3.1–§3.2)

Proposition 10.29: a running average of length k lags a linear trend by $(k - 1)/2$ and a doubly smoothed one by $k - 1$.

```

1 set.seed(9); n <- 200; tv <- 1:n; yt <- 5 + 0.3*tv + rnorm(n)
2 ma <- function(x, k) as.numeric(stats::filter(x, rep(1/k, k), sides = 1))
3 k <- 12; s1 <- ma(yt, k); s2 <- ma(s1, k); ok <- !is.na(s2)
4 c(k = k, lag_single = mean((yt - s1)[ok])/0.3, theory = (k - 1)/2,
5   lag_double = mean((yt - s2)[ok])/0.3, theory_double = k - 1,
6   trend_estimate = 2*(s1[n] - s2[n])/(k - 1), truth = 0.3)

```

```

##           k      lag_single      theory      lag_double
##      12.0000      5.4999      5.5000      11.0026
## theory_double trend_estimate      truth
##      11.0000      0.3189      0.3000

```

Theorem 10.32: exponential smoothing is the optimal one-step forecast of an $ARIMA(0, 1, 1)$, with w equal to the moving average coefficient. Three quantities that should agree are computed independently.

```

1 ew <- function(x, w) { z <- numeric(length(x)); z[1] <- x[1]

```

```

2   for (t in 2:length(x)) z[t] <- (1 - w)*x[t] + w*z[t-1]; z }
3   SSw <- function(x, w) { m <- length(x); sum((x[-1] - ew(x, w)[-m])^2) }
4   set.seed(17)
5   z <- cumsum(as.numeric(arima.sim(list(ma = -0.7), 400)))
6   im <- arima(z, order = c(0, 1, 1)); th <- -unnname(coef(im)[1])
7   c(theta_true = 0.7, theta_hat = th,
8     w_minimising_SS = optimize(function(w) SSw(z, w), c(.001, .999))$minimum,
9     sigma_arima = sqrt(im$sigma2), rms_one_step = sqrt(SSw(z, th)/399),
10    max_gap_to_arima_forecasts =
11    max(abs(c(NA, ew(z, th)[-400]) - (z - residuals(im))))[100:400]))

```

```

##                theta_true                theta_hat
##                7.000e-01                6.268e-01
##      w_minimising_SS                sigma_arima
##                6.283e-01                1.034e+00
##      rms_one_step max_gap_to_arima_forecasts
##                1.034e+00                8.882e-16

```

The smoothed series reproduces **arima**'s one-step forecasts to machine precision. On real data, Proposition 10.31(ii) is checked against **HoltWinters**, which minimises the same criterion.

```

1   t(sapply(list(Nile = as.numeric(Nile), LakeHuron = y), function(x) {

```

```

2  wst <- optimize(function(w) SSw(x, w), c(.001, .999))$minimum
3  hw <- HoltWinters(ts(x), beta = FALSE, gamma = FALSE)
4  c(w_manual = wst, w_package = 1 - unname(hw$alpha),
5    SS_manual = SSw(x, wst), SSE_package = hw$SSE,
6    SS_at_package_w = SSw(x, 1 - unname(hw$alpha))) })))

```

```

##           w_manual  w_package  SS_manual  SSE_package
## Nile      0.753443  0.75344212 2038871.83  2038871.83
## LakeHuron 0.001066  0.00006611    53.88    53.87
##           SS_at_package_w
## Nile           2038871.83
## LakeHuron           53.87

```

On the Nile the two agree to every printed digit and the optimum is interior. On Lake Huron both push w to the boundary: the series is an autoregression, not an $ARIMA(0, 1, 1)$, and the criterion prefers no smoothing at all.

5.8 Regression trees (§4)

Theorem 10.47 makes the best split of a node computable by one sweep of a sorted predictor. The result is compared with the split **tree** chooses.

```

1 suppressMessages({library(ISLR2); library(tree)})
2 best <- function(x, yv) {
3   o <- order(x); xs <- x[o]; ys <- yv[o]; m <- length(yv)
4   cs <- cumsum(ys); j <- 1:(m - 1)
5   g <- (cs[j]/j - (cs[m] - cs[j])/(m - j))^2 * j*(m - j)/m
6   g[xs[j] == xs[j + 1]] <- -Inf
7   i <- which.max(g); c(cut = (xs[i] + xs[i + 1])/2, gain = g[i]) }
8 tab <- t(sapply(setdiff(names(Boston), "medv"),
9               function(v) best(Boston[[v]], Boston$medv)))
10 round(head(tab[order(-tab[, "gain"]), ], 4), 3)

```

```

##           cut  gain
## rm        6.941 19340
## lstat      9.725 18896
## indus      6.660 11083
## ptratio   19.900 10439

```

```

1 tr <- tree(medv ~ ., data = Boston)
2 head(tr$frame[, c("var", "n", "dev", "yval")], 3)

```

```
##      var    n    dev  yval
## 1      rm 506 42716 22.53
## 2 lstat 430 17317 19.93
## 4     dis 255  6632 23.35
```

The identity (39) is verified at that split.

```
1 yv <- Boston$medv; L <- Boston$rm < 6.941; m <- length(yv)
2 d <- c(total = sum((yv - mean(yv))^2),
3       left = sum((yv[L] - mean(yv[L]))^2),
4       right = sum((yv[!L] - mean(yv[!L]))^2),
5       between = sum(L)*sum(!L)/m*(mean(yv[L]) - mean(yv[!L]))^2)
6 c(d, identity_gap = unname(d[1] - d[2] - d[3] - d[4]))
```

```
##          total          left          right          between
## 4.272e+04    1.732e+04    6.059e+03    1.934e+04
## identity_gap
## -7.276e-12
```

Proposition 10.49: monotone transformations of the predictors change nothing.

```
1 B2 <- Boston; B2$rm <- exp(B2$rm/2); B2$lstat <- log(B2$lstat)
2 tr2 <- tree(medv ~ ., data = B2)
```

```
3 c(largest_difference_in_fitted = max(abs(predict(tr) - predict(tr2))),  
4   leaves_original = sum(tr$frame$var == "<leaf>"),  
5   leaves_transformed = sum(tr2$frame$var == "<leaf>"))
```

```
## largest_difference_in_fitted      leaves_original  
##                               0                9  
##      leaves_transformed  
##                               9
```

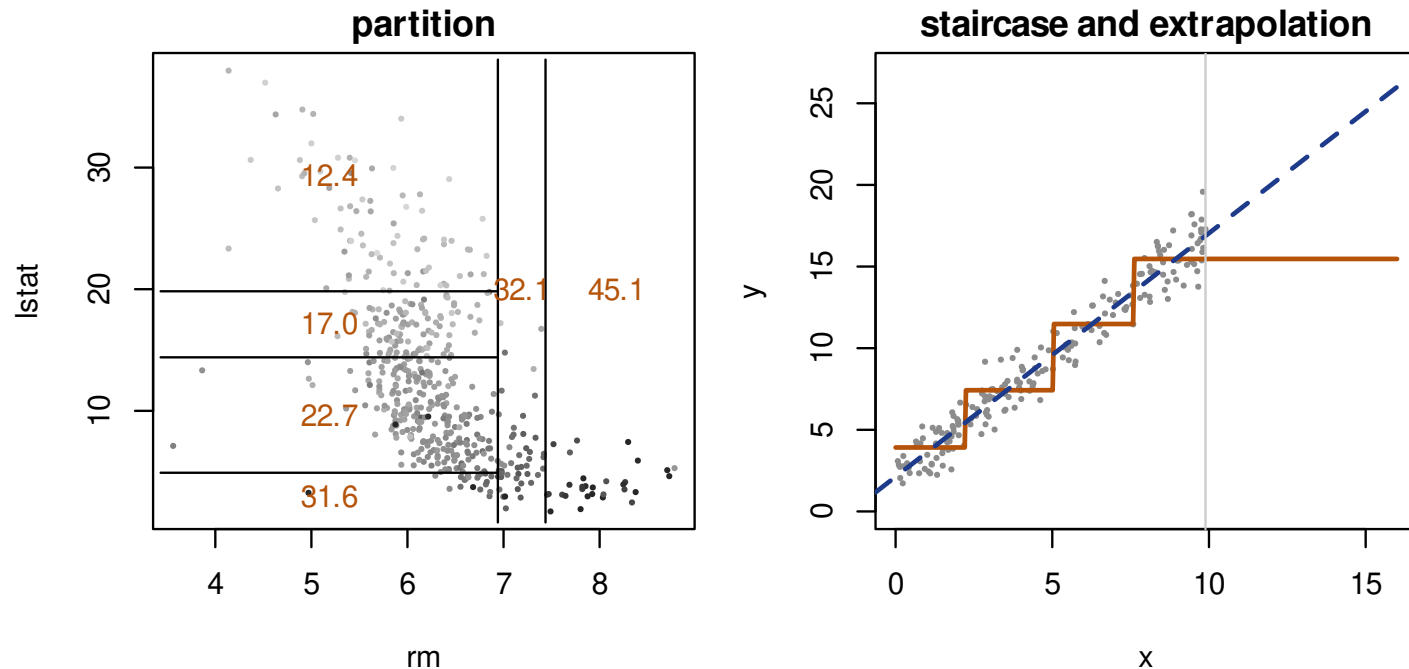


Figure 10.5: Left: the boxes a two-predictor tree cuts out of the **Boston** feature space, shading by **medv**. Right: a tree (solid) and a straight line (dashed) fitted to data generated from $y = 2 + 1.5x + \varepsilon$. The tree approximates the line by a staircase inside the data and is exactly flat outside it, which is what Definition 10.43 forces.

```

1 nd <- data.frame(xx = c(2, 5, 8, 12, 20))
2 round(rbind(tree = predict(t1, nd), linear = predict(lm(yy ~ xx, dd), nd),
3           truth = 2 + 1.5*nd$xx), 3)

```

```

##           1      2      3      4      5
## tree    3.918 7.424 15.47 15.47 15.47
## linear  5.110 9.585 14.06 20.03 31.96
## truth   5.000 9.500 14.00 20.00 32.00

```

6 Pitfalls

6.1 Fitting an autoregression to a series with a unit root

Theorem 10.16 assumes $|\beta_1| < 1$. Applied at $\beta_1 = 1$ its standard error is $\sqrt{(1 - \hat{\beta}_1^2)/T}$, which collapses towards zero, and the reported t ratio for the hypothesis $\beta_1 = 1$ has none of its usual meaning.

```

1 set.seed(4); B <- 3000
2 t(sapply(c(0.8, 0.95, 1.0), function(ph) {
3   r <- t(replicate(B, {

```

```

4      z <- if (ph == 1) cumsum(rnorm(200)) else
5          as.numeric(arima.sim(list(ar = ph), 200))
6      cf <- summary(lm(z[-1] ~ z[-200]))$coef
7      c(cf[2, 1], (cf[2, 1] - ph)/cf[2, 2]) }))
8  c(phi = ph, mean_b1 = mean(r[, 1]),
9     mean_t = mean(r[, 2]), sd_t = sd(r[, 2]),
10    reject_at_1.96 = mean(abs(r[, 2]) > 1.96)) }))

```

```

##          phi mean_b1  mean_t   sd_t reject_at_1.96
## [1,] 0.80  0.7824 -0.3102 0.9920          0.06033
## [2,] 0.95  0.9291 -0.6317 0.9655          0.09100
## [3,] 1.00  0.9741 -1.5179 0.8177          0.29200

```

Diagnostic. A fitted $\hat{\beta}_1$ close to 1, a correlogram that decays hardly at all, and a Dickey–Fuller test that does not reject. **Fix.** Test for a unit root with the right critical values (§5.5) and difference the series if it is not rejected. At $\beta_1 = 1$ the t statistic for the true value is centred well below zero and rejects far more often than 5%; at $\beta_1 = 0.95$ it is already visibly off.

6.2 Reading a partial correlogram as if the series were stationary

FREES notes that a large lag one partial autocorrelation is itself a sign of nonstationarity. The reverse mistake is to select a model order from the partial correlogram of a series that has not been differenced.

```
1 dax <- as.numeric(EuStockMarkets[, "DAX"])
2 round(rbind(level = pacf(dax, plot = FALSE, lag.max = 6)$acf[1:6],
3             log_difference = pacf(diff(log(dax)), plot = FALSE,
4                                   lag.max = 6)$acf[1:6]), 4)
```

```
##           [,1]    [,2]    [,3]    [,4]    [,5]
## level      0.9974  0.0272 -0.0077  0.0270 -0.0436
## log_difference -0.0004 -0.0267 -0.0105 -0.0004 -0.0323
##           [,6]
## level      -0.0326
## log_difference  0.0021
```

Diagnostic. A first partial autocorrelation at 0.998, with everything after it inside the band. **Fix.** Difference first, then identify. On the level the partial correlogram says $AR(1)$ with $\hat{\beta}_1 \approx 1$, which is the random walk in disguise; on the log differences it says nothing beyond lag 2.

6.3 Judging residuals of a fitted model against χ_K^2

§5.3 measured the size of the portmanteau test at 2.9% when the degrees of freedom are not reduced. Under-rejection is the safer direction, but it hides genuine misspecification. The reverse error, using K degrees of freedom for a heavily parametrised model, is not safe.

```

1 set.seed(13); B <- 2000; n <- 200; K <- 10
2 q <- replicate(B, {
3   z <- as.numeric(arima.sim(list(ar = c(0.5, -0.3, 0.2, 0.1)), n))
4   e <- residuals(arima(z, order = c(4, 0, 0), method = "CSS"))
5   rk <- acf(e, plot = FALSE, lag.max = K)$acf[2:(K+1)]
6   n*(n + 2)*sum(rk^2/(n - 1:K)) })
7 c(mean_Q = mean(q), K = K, K_minus_p = K - 4,
8   reject_at_K = mean(q > qchisq(.95, K)),
9   reject_at_K_minus_p = mean(q > qchisq(.95, K - 4)))

```

```

##              mean_Q              K              K_minus_p
##              6.1676             10.0000             6.0000
## reject_at_K reject_at_K_minus_p
##              0.0060             0.0525

```

Diagnostic. A portmanteau statistic evaluated with the same degrees of freedom whatever the

model order. **Fix.** Subtract the number of fitted autoregressive and moving average coefficients. With four of them the mean of Q_{LB} is near $K - 4$, and only the reduced degrees of freedom give a test of the advertised size.

6.4 Expecting a tree to extrapolate

```
1 set.seed(6); n <- 300
2 x1 <- runif(n, 0, 10); x2 <- runif(n, 0, 10)
3 yv <- 3 + 2*x1 - 1.5*x2 + rnorm(n)
4 d <- data.frame(x1, x2, yv)
5 nd <- data.frame(x1 = c(5, 9, 14, 30), x2 = 5)
6 tf <- tree(yv ~ x1 + x2, data = d)
7 round(rbind(tree = predict(tf, nd), linear = predict(lm(yv ~ x1 + x2, d), nd),
8           truth = 3 + 2*nd$x1 - 1.5*nd$x2), 3)
```

```
##           1      2      3      4
## tree    5.421 12.39 12.39 12.39
## linear  5.478 13.48 23.49 55.50
## truth   5.500 13.50 23.50 55.50
```

Diagnostic. Identical predictions for inputs far outside the training range, and a tree fitted to a

linear relation using many splits to trace a straight line. **Fix.** Nothing repairs this within a single tree: the predictor is constant on each box by construction (Definition 10.43). Either use a model that is linear in the offending variable, or restrict the tree to interpolation. The staircase inside the data can be reduced by ensembles (Unit 12); the flatness outside it cannot.

6.5 Reporting a GARCH fit without checking the optimiser

The likelihood (36) is not concave, and the parameters live on a constrained set. A fit started elsewhere can stop elsewhere.

```

1 starts <- rbind(c(log(1e-6), qlogis(0.07), qlogis(0.95)),
2                 c(log(1e-3), qlogis(0.40), qlogis(0.30)),
3                 c(log(1e-9), qlogis(0.01), qlogis(0.99)))
4 t(apply(starts, 1, function(p0) {
5   o <- optim(p0, nll, x = ed, method = "BFGS",
6             control = list(reltol = 1e-14, maxit = 2000))
7   c(omega = exp(o$par[1]), alpha = plogis(o$par[2]),
8     delta = plogis(o$par[3])*(1 - plogis(o$par[2])),
9     loglik = -o$value, convergence = o$convergence) }))

```

##	omega	alpha	delta	loglik	convergence
----	-------	-------	-------	--------	-------------

```
## [1,] 4.756e-06 6.845e-02 0.8876    5963    0
## [2,] 2.944e-18 4.446e-50 1.0000    5865    0
## [3,] 9.315e-09 1.363e-45 1.0000    5870    0
```

Diagnostic. Different starting values reaching different log-likelihoods, all with a convergence code of zero. **Fix.** Start from several places and keep the best, and report the log-likelihood so that the comparison can be made. Two of the three starts here stop at a degenerate point, with $\hat{\alpha}$ numerically zero, $\hat{\delta}$ at its upper bound and a log-likelihood about 95 below the best; the optimiser reports success at all three. A GARCH fit whose $\hat{\alpha} + \hat{\delta}$ is exactly 1 and whose $\hat{\omega}$ is at machine zero has not converged to anything, whatever the exit code says.

7 Exercises

Exercise 10.1. An $AR(1)$ series has $\beta_0 = 3$, $\beta_1 = 0.6$ and $\sigma^2 = 4$. Find its mean, variance, and the correlations at lags 1, 2 and 5. How large is the series' standard deviation compared with that of its innovations?

Solution. By Theorem 10.3, $\mu = 3/(1 - 0.6) = 7.5$ and $\gamma(0) = 4/(1 - 0.36) = 6.25$, so the series has standard deviation 2.5 against the innovations' 2, a ratio of $1/\sqrt{1 - \beta_1^2} = 1.25$.

The correlations are $\rho_k = 0.6^k$: 0.6, 0.36 and 0.0778. The correlogram decays geometrically and is inside a $\pm 2/\sqrt{T}$ band by lag 5 for any T below about 660, which is why an $AR(1)$ with a moderate coefficient looks like white noise after a few lags. ■

Exercise 10.2. A stationary $AR(1)$ is fitted to $T = 100$ observations and gives $\hat{\beta}_1 = 0.9$. What is the approximate standard error, and how does it change if $\hat{\beta}_1$ were 0.5? Explain the direction of the change.

Solution. By Theorem 10.16 and Corollary 10.17 the standard error is $\sqrt{(1 - \hat{\beta}_1^2)/T}$: $\sqrt{0.19/100} = 0.0436$ at 0.9 and $\sqrt{0.75/100} = 0.0866$ at 0.5, twice as large.

The reason is that the regressor is the series itself. A larger $|\beta_1|$ gives the series a larger variance, $\gamma(0) = \sigma^2/(1 - \beta_1^2)$, so the design spreads out while the error variance σ^2 stays fixed, and the slope is estimated more precisely. The limit $\beta_1 \rightarrow 1$ is not a limit of ever-better estimation of the same kind: the asymptotics break down there (§3.4). ■

Exercise 10.3. A correlogram decays geometrically with $r_1 = 0.7$ and a partial correlogram has $\hat{\alpha}_{1,1} = 0.7$ with everything after it inside the band. Which model, and why not the other one?

Solution. An $AR(1)$ with $\hat{\beta}_1 \approx 0.7$. Theorem 10.12 says an $AR(p)$ has a partial correlogram that vanishes beyond lag p , which matches at $p = 1$, and Theorem 10.3 says its correlogram decays geometrically, which also matches.

An $MA(q)$ is ruled out by Proposition 10.8: its correlogram would be exactly zero beyond lag q , and here the correlogram decays without ever cutting off. The two diagnostics are complementary and the pattern here is the autoregressive one on both. ■

Exercise 10.4. For which β_1 is $y_t = \beta_0 + \beta_1 y_{t-1} + \varepsilon_t$ stationary? Is $y_t = 1.02y_{t-1} - 0.24y_{t-2} + \varepsilon_t$ stationary, despite a coefficient exceeding one?

Solution. For $|\beta_1| < 1$ (Theorem 10.3); at $\beta_1 = 1$ the process is a random walk and no weakly stationary solution exists (Proposition 10.4).

For the second, Theorem 10.9 asks about the roots of $\Phi(x) = 1 - 1.02x + 0.24x^2$. The discriminant is $1.02^2 - 4(0.24) = 0.0804$, so the roots are $\{1.02 \pm 0.2835\}/0.48$, that is 2.71 and 1.54, both

outside the unit circle. The process is stationary. The condition is on the roots and never on the size of the coefficients: $\beta_1 + \beta_2 = 0.78 < 1$ here, which is what actually controls the persistence.

Exercise 10.5. An $AR(1)$ fit gives $\hat{\beta}_0 = 2$, $\hat{\beta}_1 = 0.8$, $s = 3$, and the last observation is $y_T = 20$. Forecast one, three and ten steps ahead with approximate 95% intervals. What do the forecast and the interval approach?

Solution. $\hat{\mu} = 2/(1 - 0.8) = 10$, so by Theorem 10.23 $\hat{y}_{T+l} = 10 + 0.8^l(20 - 10)$: 18.0, 15.12 and 11.07.

The half-widths are $2s\{\sum_{k<l} 0.8^{2k}\}^{1/2}$, that is $2(3)\sqrt{1} = 6.00$, $2(3)\sqrt{2.0496} = 8.59$ and $2(3)\sqrt{2.7458} = 9.94$, giving $(12.00, 24.00)$, $(6.53, 23.71)$ and $(1.13, 21.02)$.

Both converge: the forecast to $\hat{\mu} = 10$ and the half-width to $2s/\sqrt{1 - 0.8^2} = 2(3)/0.6 = 10$. Far enough ahead the model says only that the observation will come from the marginal distribution of the series, which is the essential difference from a random walk, whose interval grows without bound.

Exercise 10.6. An $AR(2)$ is fitted to $T = 120$ observations and $Q_{LB} = 19.4$ at $K = 12$. Is the fit adequate? What changes if the same number is quoted for the undifferenced series with no model fitted?

Solution. With an $ARMA(p, q)$ fitted the reference is $\chi^2_{K-p-q} = \chi^2_{10}$, whose 95% point is 18.31. Since $19.4 > 18.31$ the hypothesis of no residual autocorrelation is rejected at 5%: something remains, and a higher order or a moving average term is worth trying. With no model fitted the reference is χ^2_{12} by Theorem 10.21, whose 95% point is 21.03, and the same number would not be rejected. The two conclusions are opposite, which is why the degrees of freedom must be reduced by the number of fitted autoregressive and moving average coefficients; §5.3 and §6.3 measure the size of the error. ■

Exercise 10.7. Unit 9 could prove only $\mathbf{P}\{|r_k| > 2/\sqrt{T}\} \leq \frac{1}{4}$ for white noise. What is the correct limiting value, what made the improvement possible, and what does the band still not license?

Solution. By Theorem 10.18, $\sqrt{T}r_k \xrightarrow{d} N(0, 1)$, so the limiting exceedance probability is $P\{|Z| > 2\} = 0.0455$ (Corollary 10.19), which explains the 95.6% and 95.8% Unit 9 measured. What made it possible is the central limit theorem for autoregressive scores in the prerequisites: the summands $\xi_t \xi_{t-k}$ are uncorrelated but dependent, and Lindeberg–Feller needs independence. Writing them as $u_{t-1} \xi_t$ with $u_{t-1} = \xi_{t-k}$ puts them in exactly the covered form. It still does not license reading twenty lags at once. Each is an individual 4.55% test, so about one point outside the band is expected under white noise, and the joint statement is Q_{LB} , not the picture. ■

Exercise 10.8. Show that exponential smoothing with parameter w is the optimal one-step forecast of some model, and identify it. What are the forecasts two and three steps ahead?

Solution. By Theorem 10.32 the model is $ARIMA(0, 1, 1)$, $\Delta y_t = \varepsilon_t - \theta \varepsilon_{t-1}$, with $\theta = w$: the optimal one-step forecast satisfies $\hat{s}_T = (1 - \theta)y_T + \theta \hat{s}_{T-1}$, which is the smoothing recursion, and the one-step error is exactly ε_T .

The forecasts further ahead are all $\hat{s}_T$. Taking conditional expectations in $y_{T+2} = y_{T+1} + \varepsilon_{T+2} -$

$\theta\epsilon_{T+1}$ gives $\mathbf{E}[y_{T+2} \mid \mathcal{L}_T] = \mathbf{E}[y_{T+1} \mid \mathcal{L}_T] = \hat{s}_T$, and the argument repeats. The flat forecast function that looks like an assumption of the method is a consequence of the model behind it. ■

Exercise 10.9. A quarterly series with a linear trend is smoothed with a running average of length $k = 4$. By how much does the smoothed series lag the trend, and how is that repaired?

Solution. By Proposition 10.29 the expected smoothed value at t is the trend value at $t - (k - 1)/2 = t - 1.5$: it lags by one and a half quarters, so with a slope of β_1 per quarter the level is understated by $1.5\beta_1$.

Double smoothing repairs it. The doubly smoothed series lags by $k - 1 = 3$ quarters, so $2\hat{s}_T^{(1)} - \hat{s}_T^{(2)}$ is unbiased for the current level and $2(\hat{s}_T^{(1)} - \hat{s}_T^{(2)})/(k - 1)$ is unbiased for the slope. §5.7 checks both on simulated data with a known slope. ■

Exercise 10.10. A Dickey–Fuller regression on $T = 250$ observations with an intercept gives $t_{DF} = -2.10$. What is concluded, and what would a naive normal comparison have said?

Solution. The 5% point of the null distribution at this sample size is near -2.89 (§5.5), and -2.10 is well above it, so the unit root is not rejected: the series should be treated as a random walk and differenced.

A normal comparison would use -1.645 , find -2.10 beyond it, and declare the series stationary. Theorem 10.38 says why the normal is the wrong reference: at $\phi = 1$ the numerator of the estimator converges after division by T , not $\sqrt{T}$, and its limit is a recentred χ_1^2 , which is skewed and sits mostly below zero. The mistake is in the dangerous direction, since it licenses a stationary model for a series that has none. ■

Exercise 10.11. A $GARCH(1, 1)$ fit to daily returns gives $\hat{\omega} = 5 \times 10^{-6}$, $\hat{\alpha} = 0.07$ and $\hat{\delta} = 0.90$. Find the implied unconditional standard deviation. If instead an $ARCH(1)$ with $\alpha = 0.5$ were fitted, what kurtosis would the model imply?

Solution. By Theorem 10.40, $\text{var}(\varepsilon_t) = \omega/(1 - \alpha - \delta) = 5 \times 10^{-6}/0.03 = 1.667 \times 10^{-4}$, a standard deviation of 0.0129, about 1.3% per day. The persistence $\hat{\alpha} + \hat{\delta} = 0.97$ means a shock to volatility has a half life of $\ln(0.5)/\ln(0.97) = 22.8$ days.

For the $ARCH(1)$, Theorem 10.41 gives kurtosis $3(1 - \alpha^2)/(1 - 3\alpha^2) = 3(0.75)/(0.25) = 9$, three

times the normal value, even though the model is conditionally normal at every date. That is the point of the model: fat tails are produced by mixing normals of varying variance, not by assuming a fat-tailed innovation. ■

Exercise 10.12. A node holds $n = 100$ observations with $\sum (y_i - \bar{y})^2 = 800$. A candidate split sends 40 observations with mean 12 to the left and 60 with mean 18 to the right. What is the residual sum of squares after the split, and which of two candidate splits with the same mean difference should be preferred?

Solution. By Theorem 10.47 the gain is $n_L n_R (\bar{y}_L - \bar{y}_R)^2 / n = 40(60)(6)^2 / 100 = 864$. That exceeds 800, which is impossible, so no such node exists: the identity (39) bounds the gain by the node's total sum of squares. With a mean difference of 4 instead, the gain is $40(60)(16) / 100 = 384$ and the remaining sum of squares is $800 - 384 = 416$.

Between two splits with the same mean difference, prefer the more balanced one: the weight $n_L n_R / n$ is maximised at $n_L = n_R = n/2$ and falls to almost nothing when one side holds a handful of observations. That is what stops a single extreme point from being split off. ■

Exercise 10.13. A regression tree and a linear model are both fitted with `lstat` as a predictor. The analyst replaces `lstat` by $\ln(\text{lstat})$. What happens to each fit, and what happens if a test point lies beyond the training range?

Solution. The tree is unchanged: by Proposition 10.49 a strictly increasing transformation preserves the ordering of the observed values, and the split criterion depends on nothing else. §5.8 verifies that the fitted values agree exactly. The linear model changes completely, since it fits a straight line in whichever coordinate is supplied.

Beyond the training range the tree returns the mean of the outermost box, the same value for every such point, however far out. The linear model extrapolates its line, which may be wrong but at least moves. §6.4 shows a tree predicting the same number at $x = 14$ and $x = 30$ when the truth differs by 32. ■

Exercise 10.14. In R: (a) verify that conditional least squares, `arima` with the conditional sum of squares method, and the closed form agree; (b) verify that the asymptotic standard error $\sqrt{(1 - \hat{\beta}_1^2)/T}$ matches the one a regression prints; (c) show by simulation that the Ljung–Box

statistic on residuals of a fitted $AR(1)$ needs one fewer degree of freedom.

Solution. (a) and (b) are §5.2, where the two standard errors agree in the fifth decimal; (c) is §5.3.

The point to watch in (c) is that the error is one of size, not of consistency: using χ_K^2 makes the test too conservative, so a model that fails is not detected. The diagnosis is the mean of the statistic, which sits at $K - 1$ rather than K over the replicates, and §6.3 repeats it with four fitted coefficients, where the mean drops to $K - 4$. ■